G. N. KHALSA COLLEGE

AUTONOMOUS

(UNIVERSITY OF MUMBAI)

MUMBAI-400019

DEPARTMENT OF COMPUTER SCIENCE

CERTIFICATE

Exam Seat No._____

CERTIFIED that the practicals, assignment duly signed, were performed by

Mr./Ms. _____Roll No ---- of M.Sc Part I/Part II class in the Computer Science Laboratory of G. N. Khalsa College, Mumbai during academic year 20--- 20----.

He/She has completed the course of Laboratory assignments in Computer Science as contained in the course prescribed by the University of Mumbai

Sign. Of the Student

Date_____

Head of Dept.

Computer Science

Date_____

Professor-in-charge

1)_____

Date_____

2)_____

Date_____

Sign of Examiner's

1)_____

Date_____

2)_____

Date_____

Algorithm for Optimization

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Practical No: 01

Aim: Implement Contour Plots

Step 1: Install PlotlyJS in Julia

Open Julia REPL, press] to enter package manager, then:

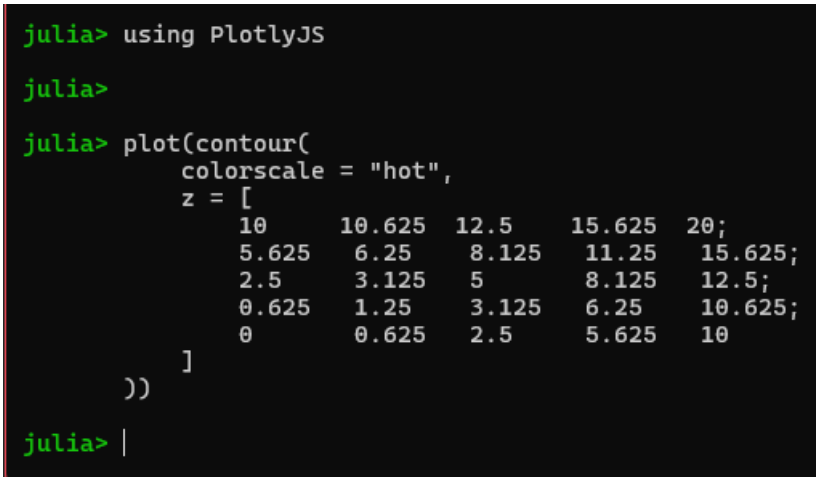
add PlotlyJS

Press Backspace to exit package manager.

Step 2: Write the Code

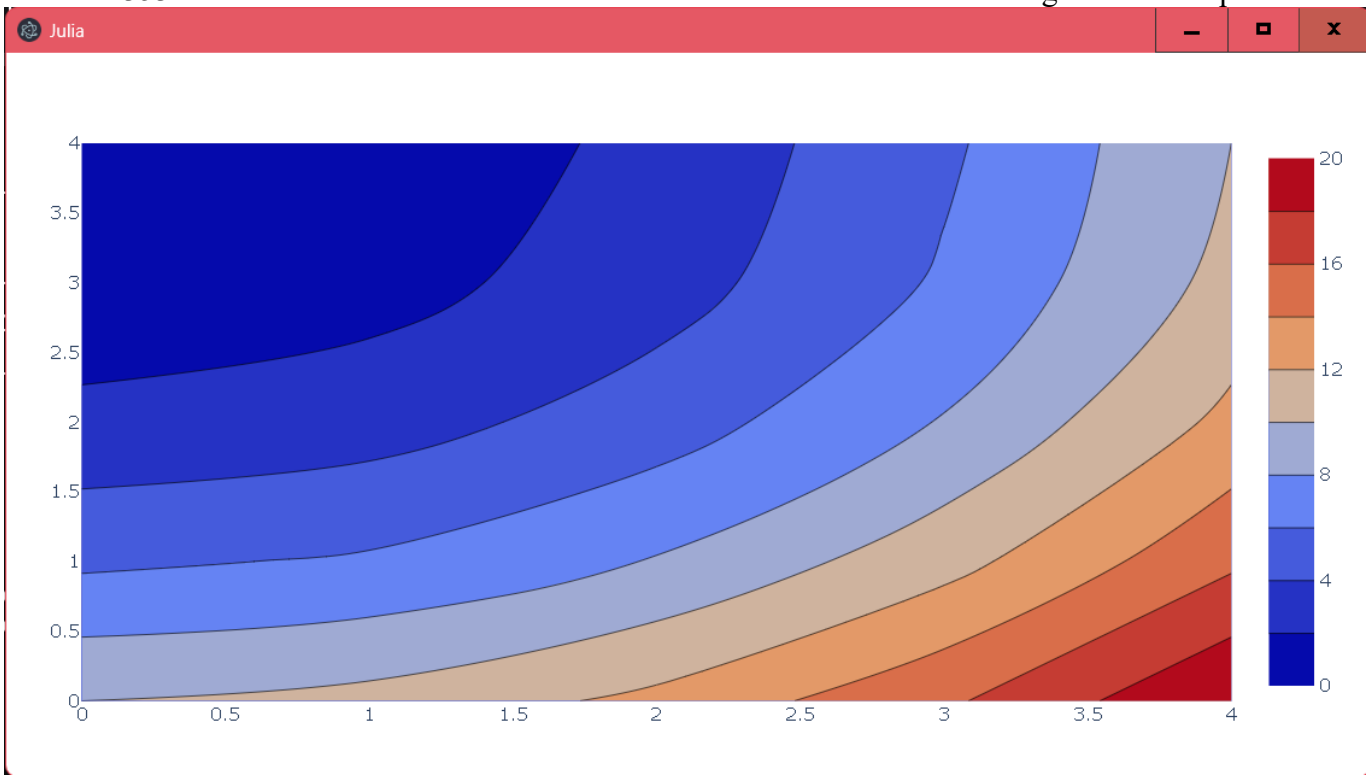
```
using PlotlyJS

plot(contour(
   colorscale = "hot",
   z = [
      10     10.625  12.5    15.625  20;
      5.625   6.25    8.125   11.25   15.625;
      2.5     3.125   5       8.125   12.5;
      0.625   1.25    3.125   6.25    10.625;
      0       0.625   2.5     5.625   10
   ]
))
```



Step 3: Run the Code

Press Enter in Julia REPL. A browser window opens with the contour plot.


Julia
4
3.5
3
2.5
2
1.5
1
0.5
0
0
0.5
1
1.5
2
2.5
3
3.5
4
20
16
12
8
4
0

Practical No: 02

Aim: **Path Finding Using Ant Colony Optimization**

Step 1: Install Julia Package

Open Julia REPL, press] to enter package manager:

add AntColony

Step 2: Write the Code (Fixed)

```
using AntColony
import AntColony: aco
# Set seed for reproducibility
using Random
Random.seed!(42)
# Create a random 10x10 distance matrix
distance_matrix = rand(10, 10)
# Symmetrize (distance from i to j = distance from j to i)
distance_matrix = (distance_matrix + distance_matrix') / 2
# Set diagonal to 0 (distance from node to itself = 0)
for i in 1:10
    distance_matrix[i, i] = 0
end
# Run ACO to find full tour
tour = aco(distance_matrix, is_tour = true)
# Run ACO to find path from node 1 to node 5
path_1_to_5 = aco(distance_matrix, start_node = 1, end_node = 5)
# Run ACO to find path from node 2 to node 10
path_2_to_10 = aco(distance_matrix, start_node = 2, end_node = 10)
println("Tour: ", tour)
println("Path 1->5: ", path_1_to_5)
println("Path 2->10: ", path_2_to_10)
```

```
julia> using AntColony

julia> import AntColony: aco

julia>

julia> # Set seed for reproducibility

julia> using Random

julia> Random.seed!(42)
TaskLocalRNG()

julia>

julia> # Create a random 10x10 distance matrix

julia> distance_matrix = rand(10, 10)
10×10 Matrix{Float64}:
```

```
julia> # Set diagonal to 0 (distance from node to itself = 0)

julia> for i in 1:10
           distance_matrix[i, i] = 0
       end

julia>

julia> # Run ACO to find full tour

julia> tour = aco(distance_matrix, is_tour = true)
10-element Vector{Int64}:
  1
  4
  8
  9
 10
  2
  5
  7
  3
  6

julia>

julia> # Run ACO to find path from node 1 to node 5

julia> path_1_to_5 = aco(distance_matrix, start_node = 1, end_node = 5)
10-element Vector{Int64}:
  1
  4
  8
  9
  7
  3
  6
 10
  2
  5
```

Step 3: Run the Code

Press Enter in Julia REPL.

```
julia> println("Tour: ", tour)
Tour: [1, 4, 8, 9, 10, 2, 5, 7, 3, 6]

julia> println("Path 1->5: ", path_1_to_5)
Path 1->5: [1, 4, 8, 9, 7, 3, 6, 10, 2, 5]

julia> println("Path 2->10: ", path_2_to_10)
Path 2->10: [2, 5, 8, 4, 1, 6, 3, 7, 9, 10]
```

Practical No: 03

Aim: **Implement Gradient Descent**

Code:

```
function gradient_descent(P, q, x0; α=0.1, maxiter=1000, ε=1e-5)
    x = copy(x0)
    ∇f(x) = P * x + q
    for i in 1:maxiter
        Δx = -∇f(x)
        norm(Δx) < ε && break
        x += α * Δx
    end
    return x
end

# Example
P = [10.0 -1.0; -1.0 1.0]
q = [0.0; -10.0]
x0 = zeros(2)
gradient_descent(P, q, x0)
```

```
julia> function gradient_descent(P, q, x0; α=0.1, maxiter=1000, ε=1e-5)
           x = copy(x0)
           ∇f(x) = P * x + q
           for i in 1:maxiter
               Δx = -∇f(x)
               norm(Δx) < ε && break
               x += α * Δx
           end
           return x
       end
gradient_descent (generic function with 1 method)

julia>

julia> # Example

julia> P = [10.0 -1.0; -1.0 1.0]
2×2 Matrix{Float64}:
 10.0  -1.0
 -1.0   1.0

julia> q = [0.0; -10.0]
2-element Vector{Float64}:
   0.0
 -10.0

julia> x0 = zeros(2)
2-element Vector{Float64}:
 0.0
 0.0

julia> gradient_descent(P, q, x0)
```

Output:

```
julia> gradient_descent(P, q, x0)
2-element Vector{Float64}:
  1.1111099828499227
 11.111100832908669
```

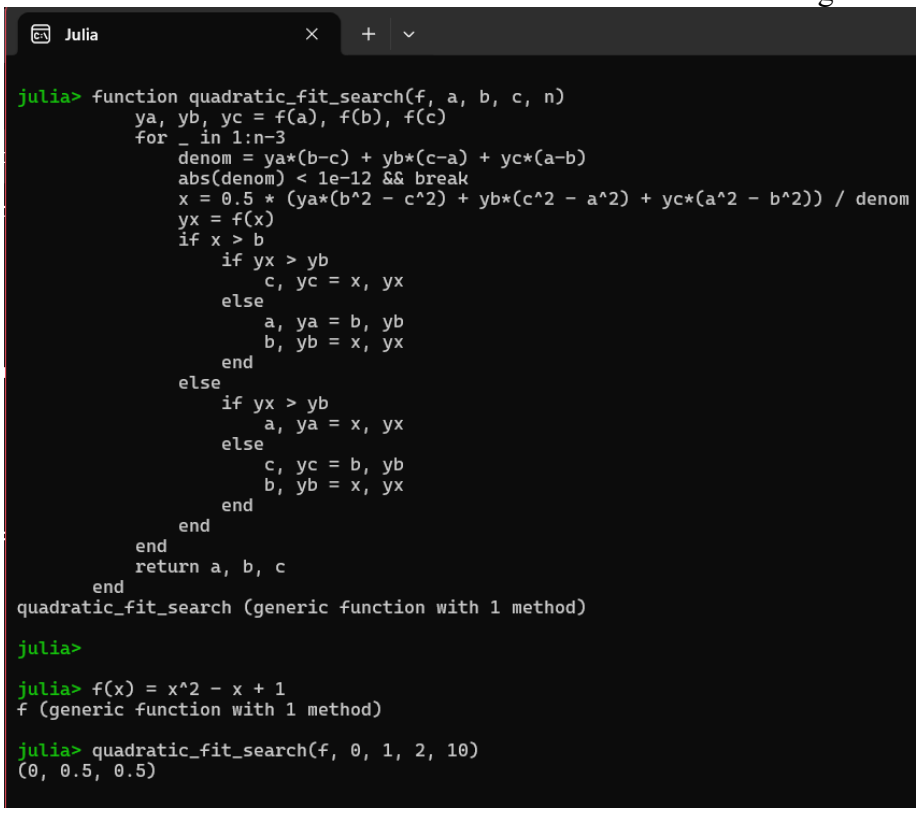
Practical No: 04

Aim: **Implement Quadratic Fit Search**

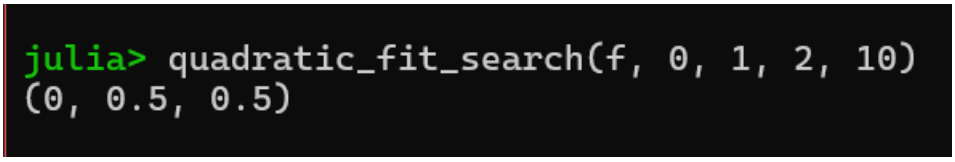
Code:

```
function quadratic_fit_search(f, a, b, c, n)
    ya, yb, yc = f(a), f(b), f(c)
    for _ in 1:n-3
        denom = ya*(b-c) + yb*(c-a) + yc*(a-b)
        abs(denom) < 1e-12 && break
        x = 0.5 * (ya*(b^2 - c^2) + yb*(c^2 - a^2) + yc*(a^2 - b^2)) / denom
        yx = f(x)
        if x > b
            if yx > yb
                c, yc = x, yx
            else
                a, ya = b, yb
                b, yb = x, yx
            end
        else
            if yx > yb
                a, ya = x, yx
            else
                c, yc = b, yb
                b, yb = x, yx
            end
        end
    end
    return a, b, c
end

f(x) = x^2 - x + 1
quadratic_fit_search(f, 0, 1, 2, 10)
```


Julia
julia> function quadratic_fit_search(f, a, b, c, n)
ya, yb, yc = f(a), f(b), f(c)
for _ in 1:n-3
denom = ya*(b-c) + yb*(c-a) + yc*(a-b)
abs(denom) < 1e-12 && break
x = 0.5 * (ya*(b^2 - c^2) + yb*(c^2 - a^2) + yc*(a^2 - b^2)) / denom
yx = f(x)
if x > b
if yx > yb
c, yc = x, yx
else
a, ya = b, yb
b, yb = x, yx
end
else
if yx > yb
a, ya = x, yx
else
c, yc = b, yb
b, yb = x, yx
end
end
end
return a, b, c
end
quadratic_fit_search (generic function with 1 method)
julia>
julia> f(x) = x^2 - x + 1
f (generic function with 1 method)
julia> quadratic_fit_search(f, 0, 1, 2, 10)
(0, 0.5, 0.5)

Output:


julia> quadratic_fit_search(f, 0, 1, 2, 10)
(0, 0.5, 0.5)

Practical No: 05

Aim: Implement Fibonacci search

Code:

```
function fibonacci_search(f, a, b, n)
   ρ = (sqrt(5) - 1) / 2            # ≈ 0.618
   x1 = a + (1 - ρ) * (b - a)       # LEFT point  (~0.382)
   x2 = a + ρ * (b - a)             # RIGHT point (~0.618)
   for i in 1:n-1
      if f(x1) < f(x2)
         b = x2
         x2 = x1
         x1 = a + (1 - ρ) * (b - a)
      else
         a = x1
         x1 = x2
         x2 = a + ρ * (b - a)
      end
   end
   return a, b
end

f(x) = x^2 - x + 1
fibonacci_search(f, 0, 2, 10)
```

```
julia> function fibonacci_search(f, a, b, n)
           ρ = (sqrt(5) - 1) / 2            # ≈ 0.618
           x1 = a + (1 - ρ) * (b - a)       # LEFT point  (~0.382)
           x2 = a + ρ * (b - a)             # RIGHT point (~0.618)
           for i in 1:n-1
               if f(x1) < f(x2)
                   b = x2
                   x2 = x1
                   x1 = a + (1 - ρ) * (b - a)
               else
                   a = x1
                   x1 = x2
                   x2 = a + ρ * (b - a)
               end
           end
           return a, b
       end
fibonacci_search (generic function with 3 methods)

julia> f(x) = x^2 - x + 1
f (generic function with 1 method)
```

Output:

```
julia> fibonacci_search(f, 0, 2, 10)
(0.48839719251114605, 0.5147084275039957)

julia>
```

Software Defined Network

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Practical No: 01

Aim: **Basic Commands using Cisco Packet Tracer**

Hardware / Devices Required

Sr No	Device	Quantity
1	Router	1
2	Switch	1
3	End Device (PC)	2
4	Connection Wire	3

Network Topology

A basic LAN consists of end devices (PCs) connected to a switch, which is further connected to a router. The router acts as the default gateway, allowing communication between different networks. In this practical, all devices are on the same subnet, so the router mainly serves as the gateway.

Command-Line Utilities

Command	Purpose	Protocol / Concept
ipconfig	Displays IP configuration of the host	Local IP settings
ping	Tests reachability between two hosts	ICMP Echo Request/Reply
tracert	Traces the path packets take to a destination	ICMP with increasing TTL
nslookup	Queries DNS to resolve names to IPs	DNS (UDP port 53)
netstat	Shows active connections, listening ports	TCP/IP stack

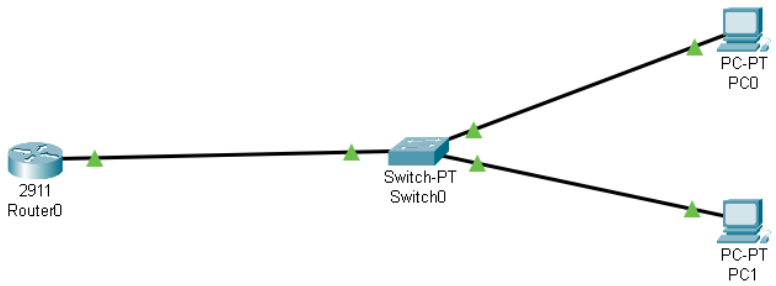
Command	Purpose	Protocol / Concept
arp -a	Shows ARP cache (IP ↔ MAC mappings)	ARP

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router1	Gi0/0	192.168.1.1	255.255.255.0	—
PC0	NIC	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.3	255.255.255.0	192.168.1.1

Step 1: Build Topology

1. Open Cisco Packet Tracer.
2. Drag and drop Router, Switch, PC0, and PC1.
3. Connect all the devices



Step 2: Configure Router

Manually Assign the Addresses

OR

Click Router1 → CLI tab → Enter the following commands:

Router> enable

Router# configure terminal

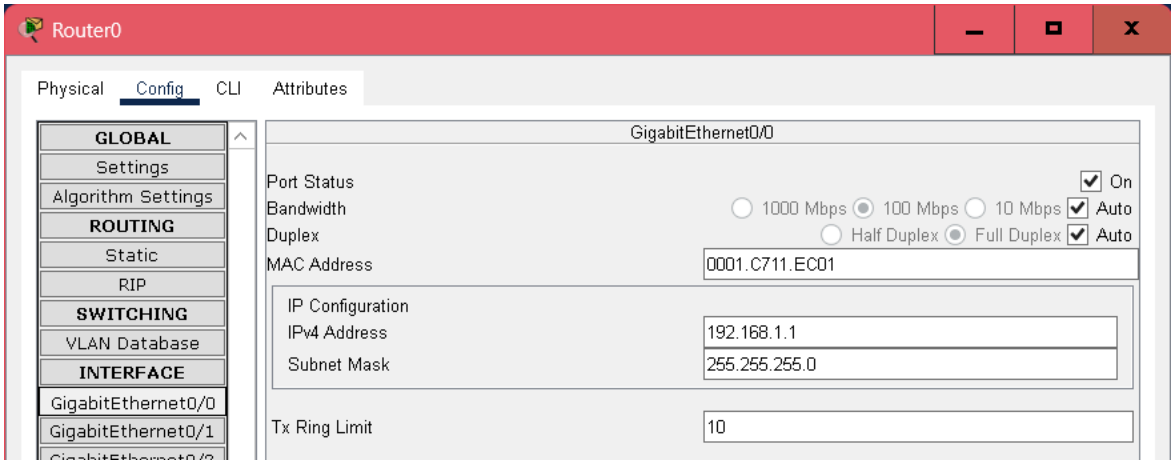
Router(config)# interface GigabitEthernet0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

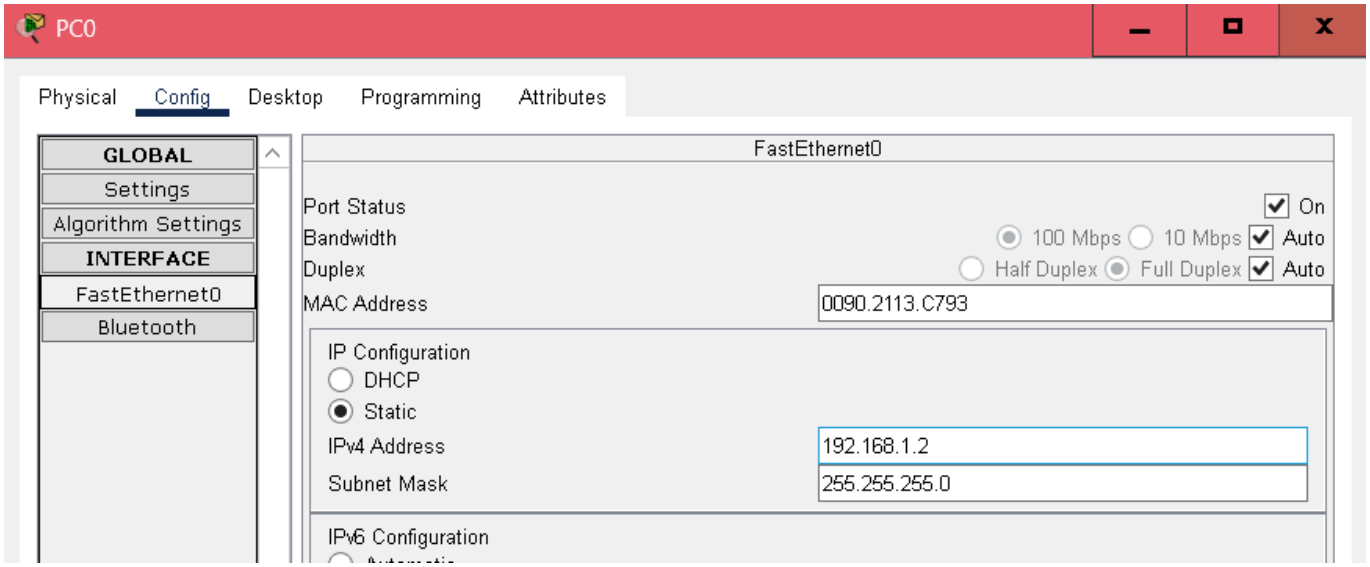
Router(config)# exit



Step 3: Configure PC0

Click PC0 → Desktop → IP Configuration:

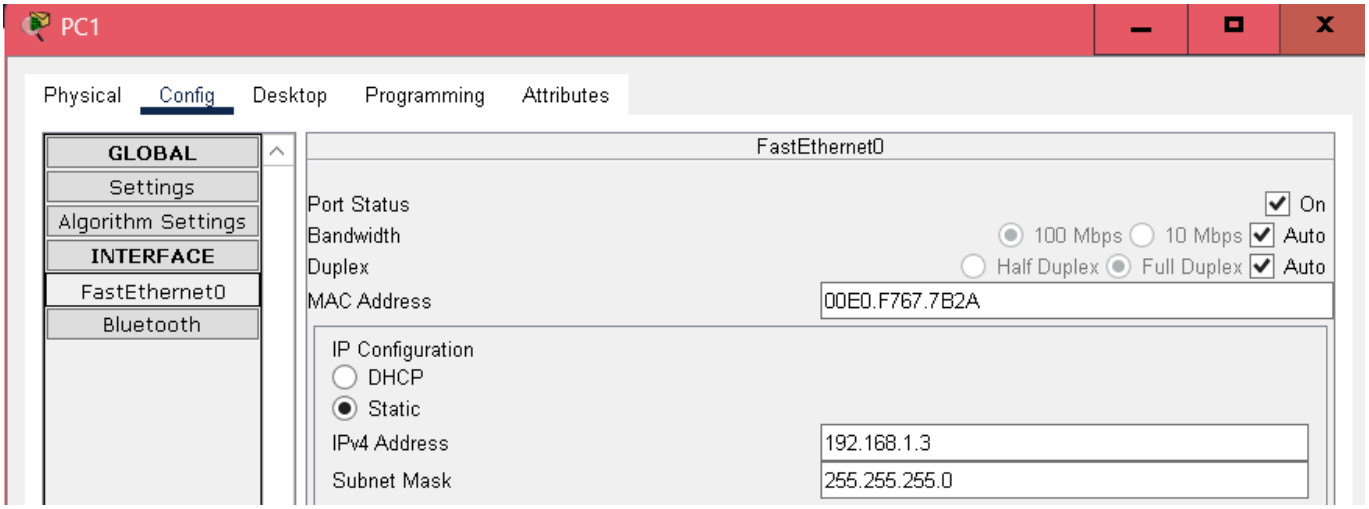
- IPv4 Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1



Step 4: Configure PC1

Click PC1 → Desktop → IP Configuration:

- IPv4 Address: 192.168.1.3
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

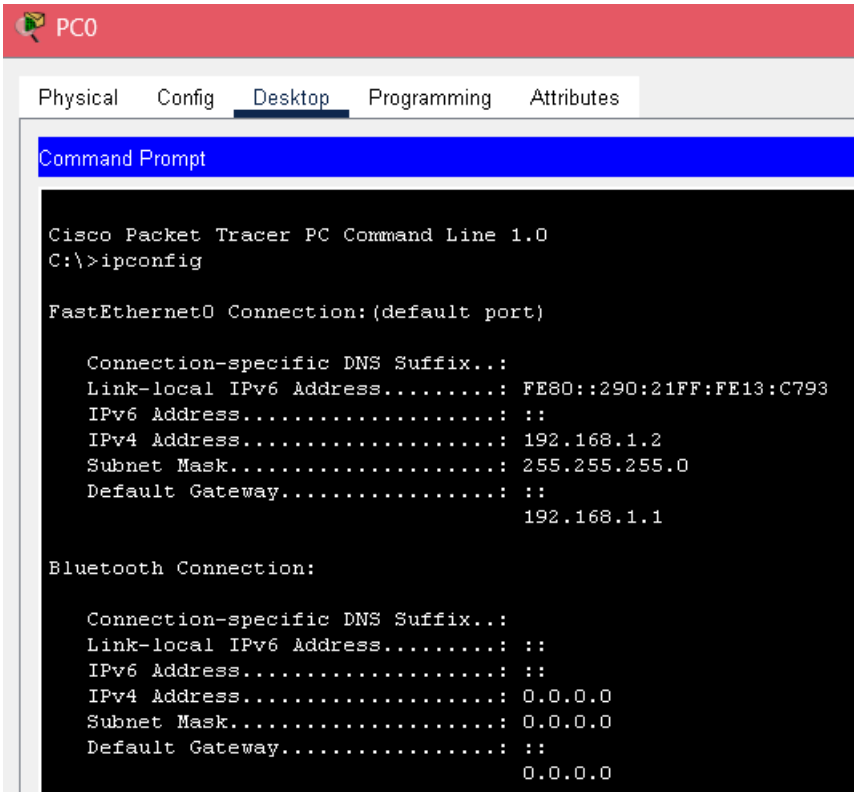


Step 5: Verify Connectivity from PC0

Open PC0 → Desktop → Command Prompt.

5.1 ipconfig

PC> ipconfig



5.2 ping

PC> ping 192.168.1.3

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

5.3 tracert

PC> tracert 192.168.1.3

```
C:\>tracert 192.168.1.3

Tracing route to 192.168.1.3 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.3

Trace complete.
```

5.4 nslookup

PC> nslookup

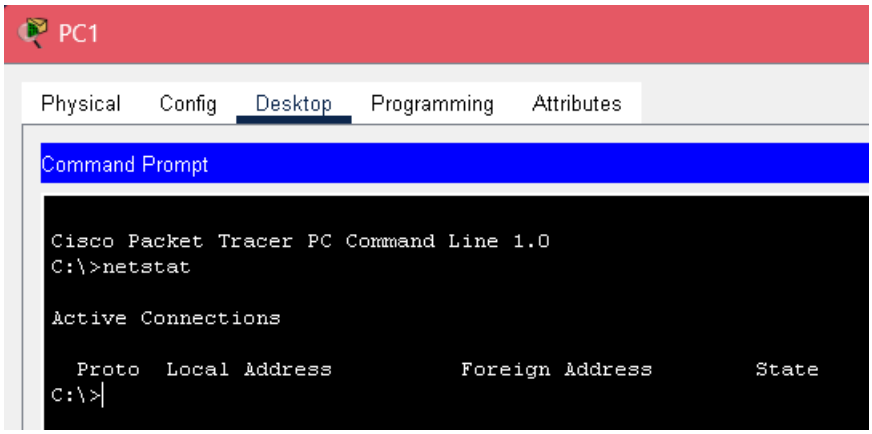
```
C:\>nslookup

Server: [255.255.255.255]
Address:   255.255.255.255

>
```

5.5 netstat

PC> netstat



Practical No: 2a

Aim: **Perform Packet Sniffing**

What is Packet Sniffing?

Packet sniffing is the process of capturing and inspecting data packets flowing across a network. It is used for:

- Network troubleshooting
- Security analysis
- Protocol debugging
- Performance monitoring

What is Wireshark?

Wireshark is a free, open-source network protocol analyzer. It captures live traffic from a network interface and displays it in a human-readable format. It supports hundreds of protocols and provides powerful display filters.

Common Display Filter Syntax

Filter	Meaning
ip.addr == 1.1.1.1	Any packet with source or destination IP 1.1.1.1
ip.src == 1.1.1.1	Packets originating from 1.1.1.1
ip.dst == 1.1.1.1	Packets destined to 1.1.1.1
http	HTTP protocol packets
http.request	HTTP request packets only
http.response.code == 200	HTTP 200 OK responses
tcp	TCP protocol packets
tcp.port == 80	TCP packets on port 80
tcp.port == 443	TCP packets on port 443

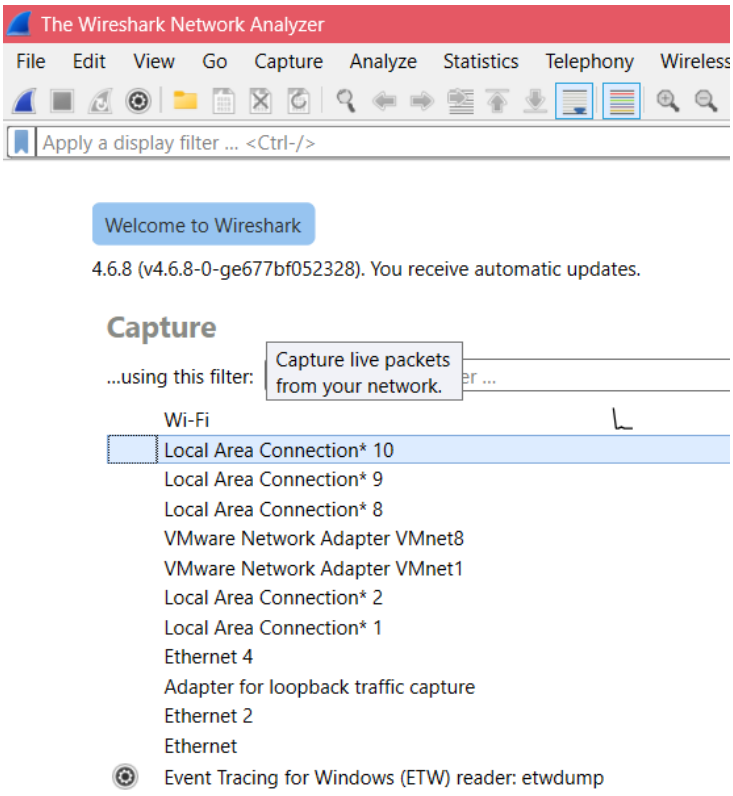
Filter	Meaning
tcp.port == 80 || tcp.port == 443	TCP packets on port 80 or 443

Step 1: Setup

1. Install Wireshark on your machine.
2. Connect to a network (Wi-Fi or Ethernet).
3. Open Wireshark.

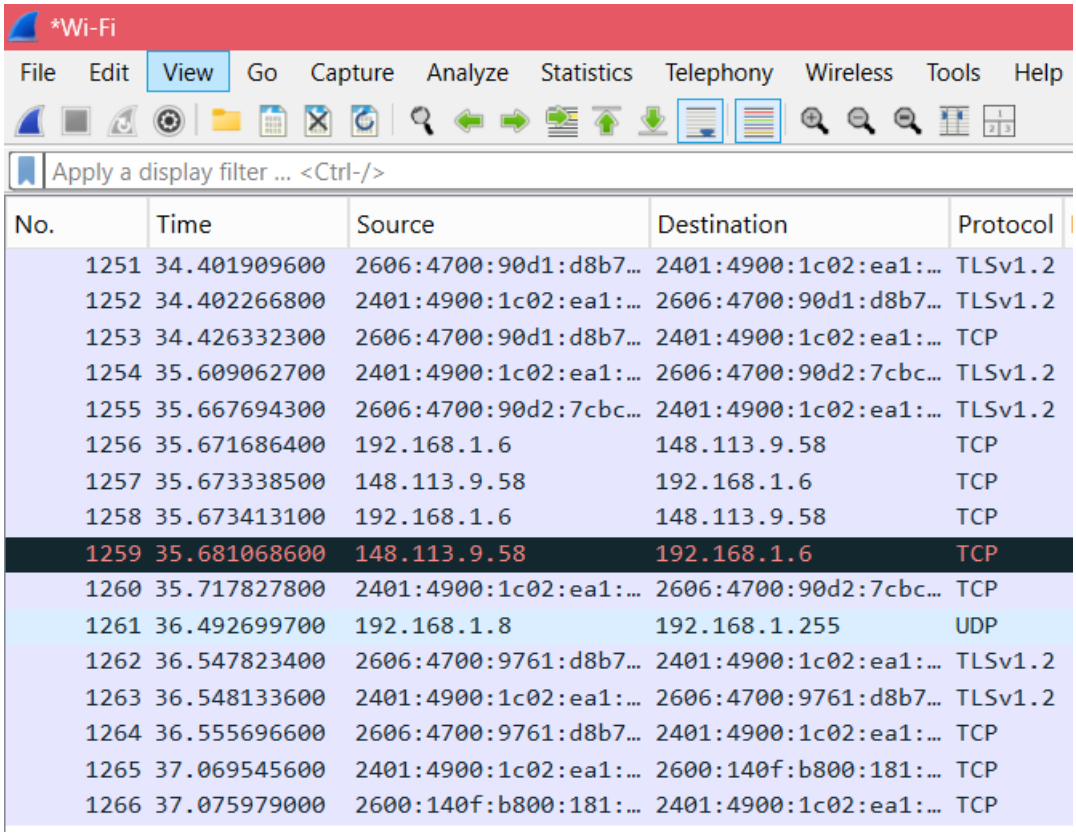
Step 2: Select Interface

- Choose the active network interface (e.g., Wi-Fi or Ethernet).
- Click the blue shark fin icon to start capturing.



Step 3: Generate Traffic

- Open a browser and visit a website (e.g., http://example.com or https://google.com).
- Run ping 1.1.1.1 in a terminal.
- This generates ICMP, DNS, TCP, HTTP/HTTPS traffic.
- To generate HTTP traffic use http://neverssl.com

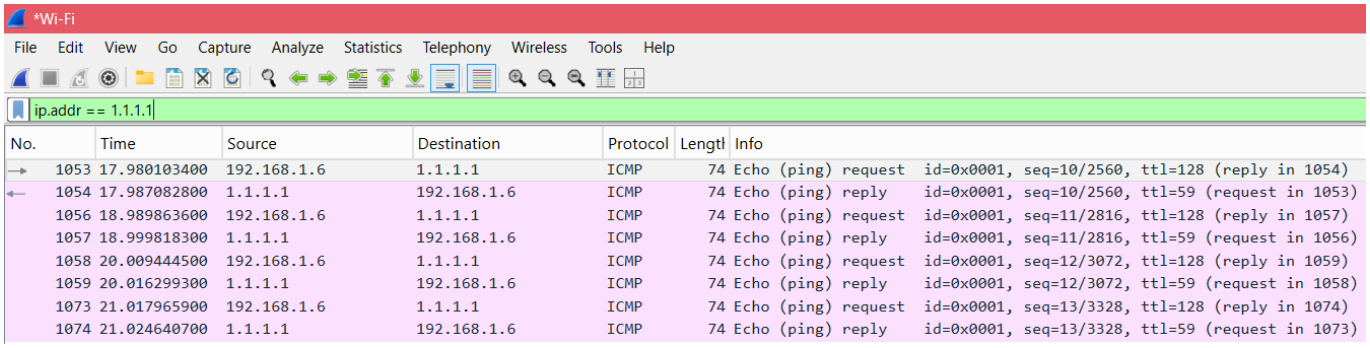


Step 4: Apply Display Filters

Filter 1: Display packets based on specific IP address

ip.addr == 1.1.1.1

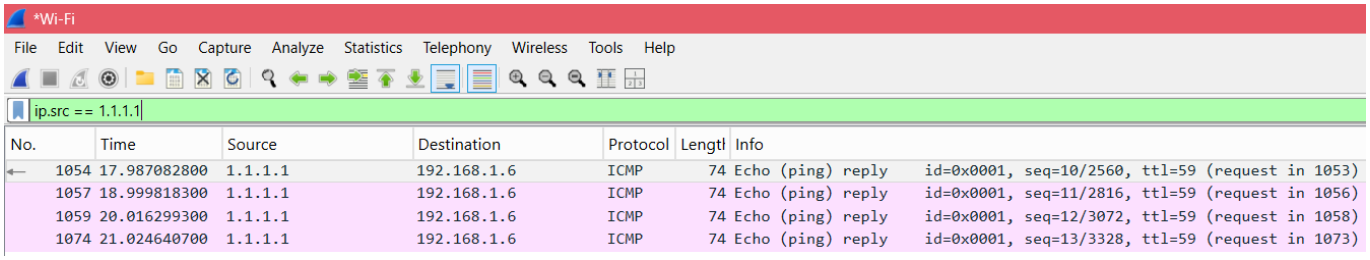
- Shows all packets where 1.1.1.1 is either source or destination.



Filter 2: Display packets coming from a specific IP address

ip.src == 1.1.1.1

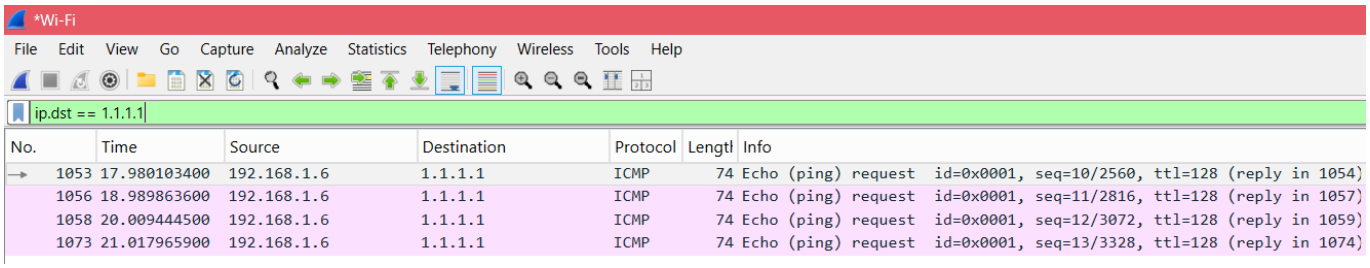
- Shows only packets originating from 1.1.1.1.



Filter 3: Display packets going to a specific IP address

ip.dst == 1.1.1.1

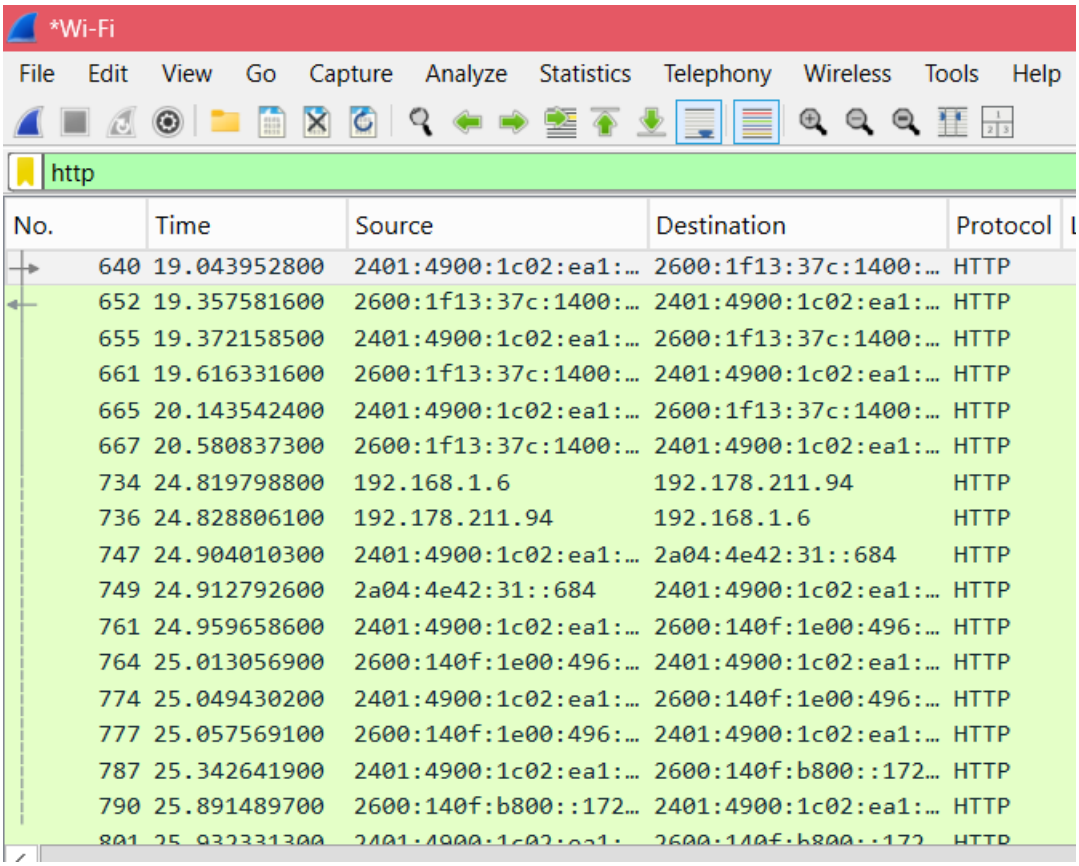
- Shows only packets destined to 1.1.1.1.



Filter 4: Display packets using HTTP protocol

http

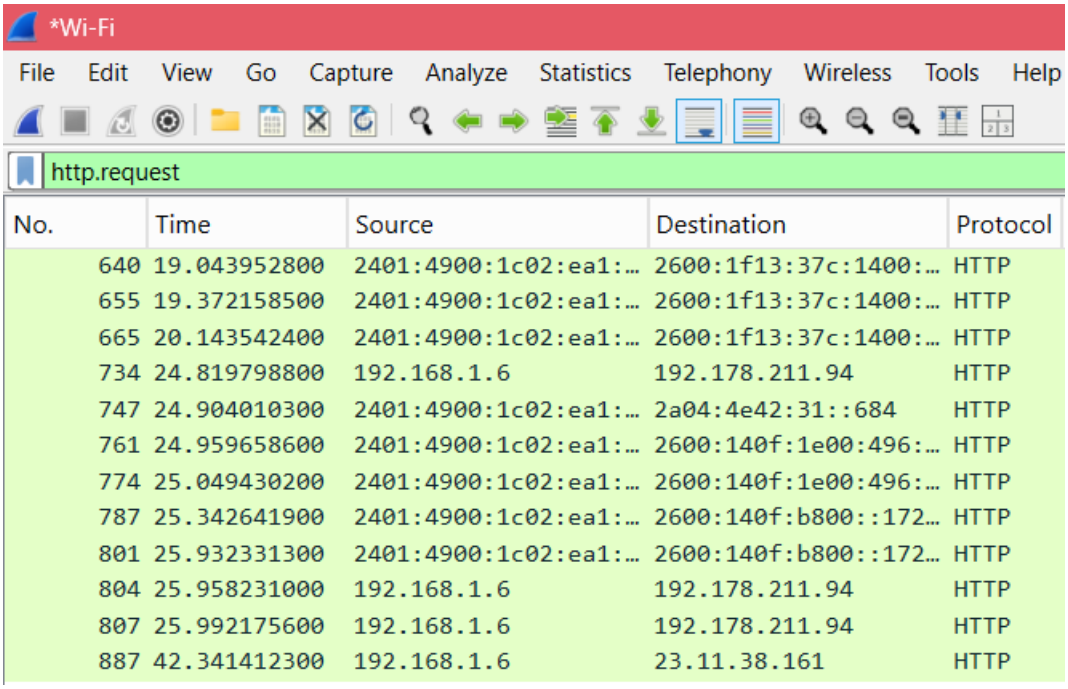
- Shows all HTTP traffic.



Filter 5: Display HTTP request packets

http.request

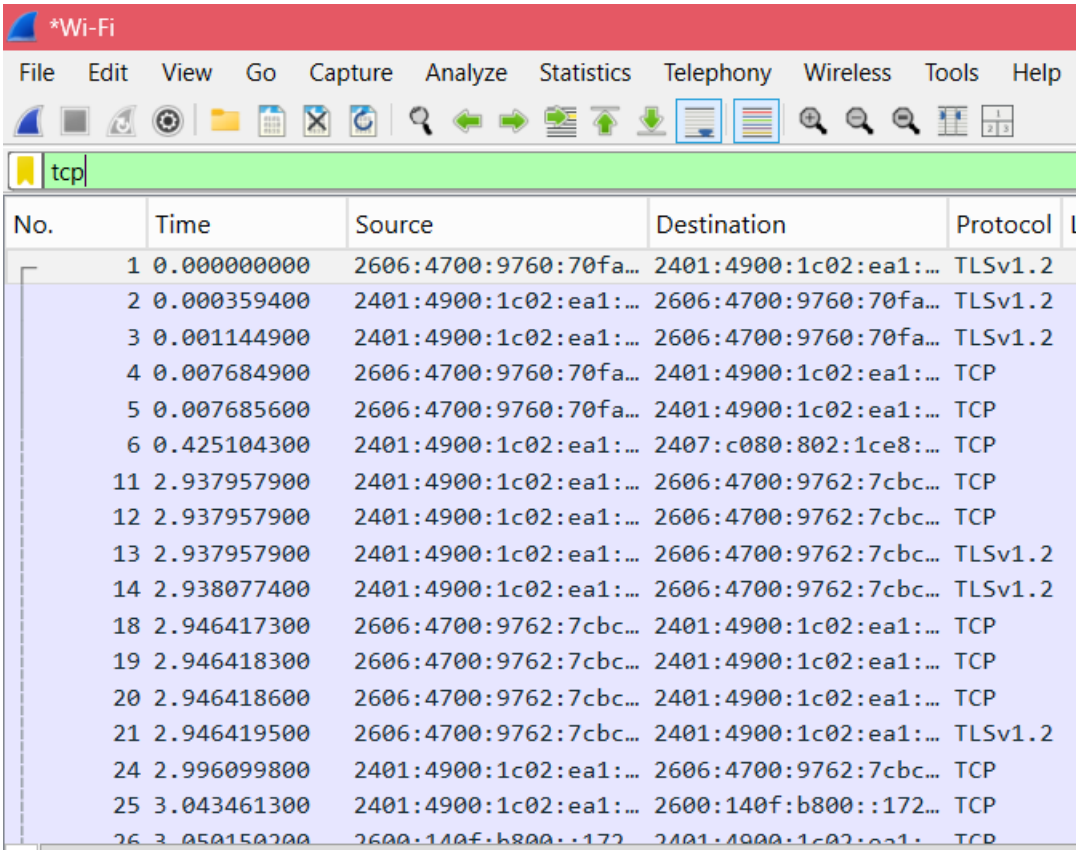
- Shows only HTTP requests (GET, POST, etc.).



Filter 6: Display TCP protocol packets

tcp

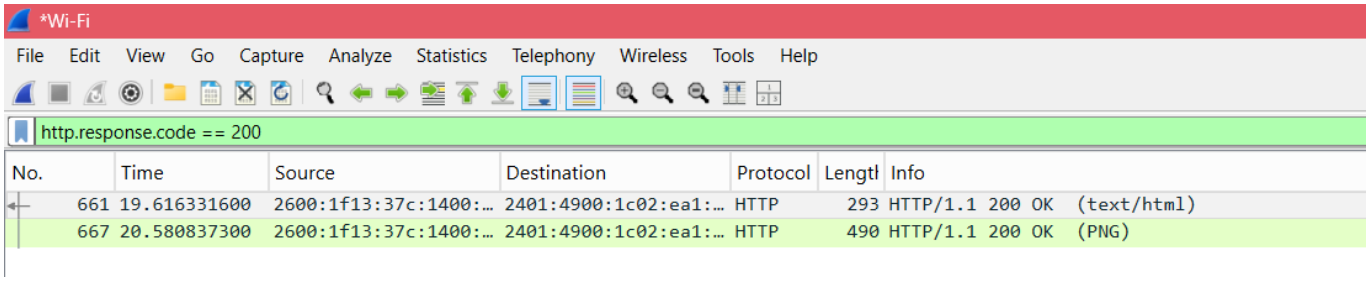
- Shows all TCP traffic.



Filter 7: Display HTTP response code 200

http.response.code == 200

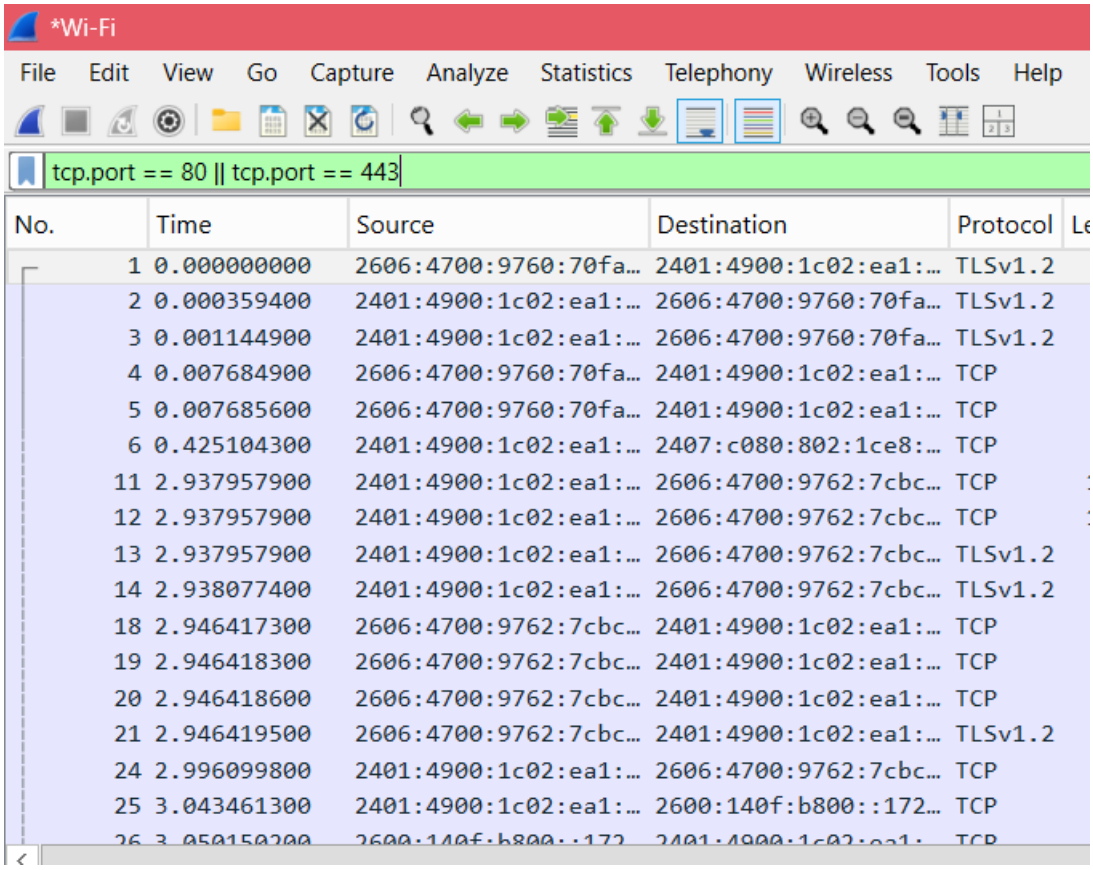
- Shows only successful HTTP responses.



Filter 8: Display packets on port 80 or 443

tcp.port == 80 || tcp.port == 443

- Shows HTTP and HTTPS traffic.



Practical No: 2b

Aim: **Integrating Wireshark with Another Tool (Nmap)**

What is Nmap?

Nmap (Network Mapper) is a free, open-source tool for network discovery and security auditing. It is used to:

- Discover hosts on a network
- Identify open ports and services
- Detect operating systems and versions
- Perform vulnerability scanning

Why Integrate Wireshark with Nmap?

- Nmap tells you **what** is open.
- Wireshark shows you **how** the scan actually happened at packet level.
- Together, they prove:
  - The scan packets were sent.
  - The target responded.
  - The TCP handshake behavior (SYN → SYN-ACK → RST).
  - The difference between open and closed ports.

TCP Flags

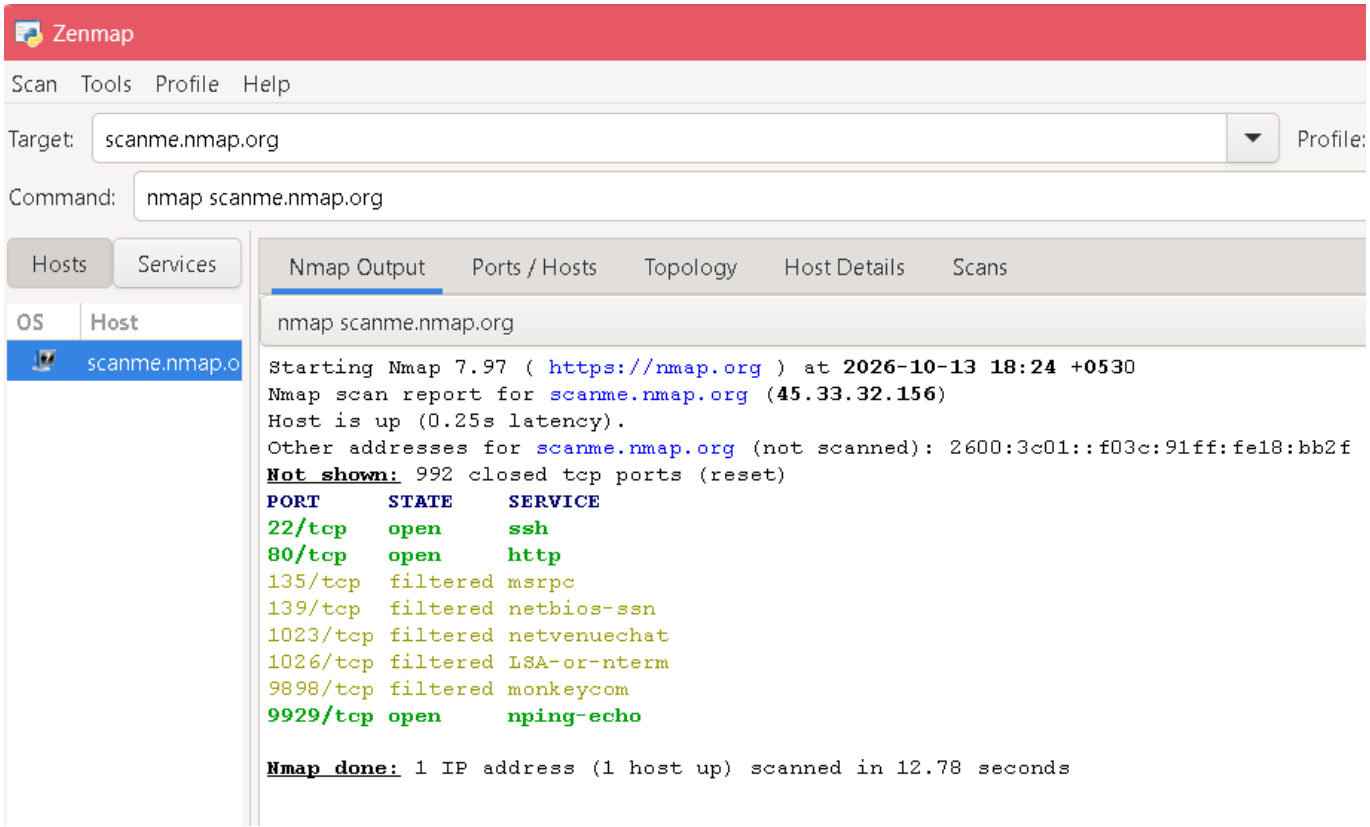
Flag	Meaning
SYN	**Synchronize — initiate connection**
ACK	**Acknowledge — confirm receipt**
RST	**Reset — abort connection**
FIN	**Finish — graceful close**
PSH	**Push — deliver data immediately**
URG	**Urgent — priority data**

Step 1: Start Wireshark Capture

1. Open Wireshark.
2. Select the active network adapter (e.g., Wi-Fi or Ethernet).
3. Click the blue shark fin icon to start capturing.
4. Traffic will start populating in the live view.

Step 2: Identify the Target

- Use scanme.nmap.org — this is a **legally authorized** target provided by Nmap for testing.
- Note its IP address (e.g., 45.33.32.156)



Important: Only scan hosts you own or have permission to scan. scanme.nmap.org is explicitly provided for this purpose.

Step 3: Execute TCP SYN Scan

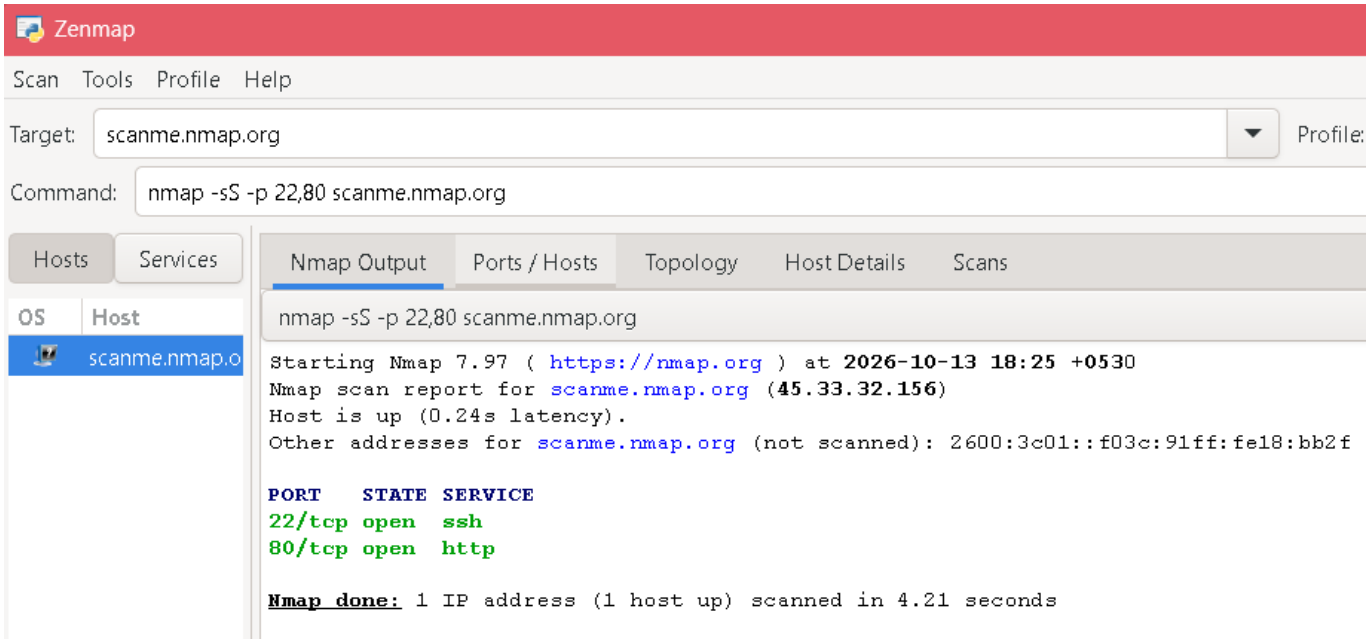
1. Open Command Prompt **as Administrator**.
2. Navigate to Nmap installation directory:

cd C:\Program Files (x86)\Nmap

OR

2. Use Nmap GUI Application
3. Run the SYN scan on ports 22 (SSH) and 80 (HTTP):

nmap -sS scanme.nmap.org -p 22,80



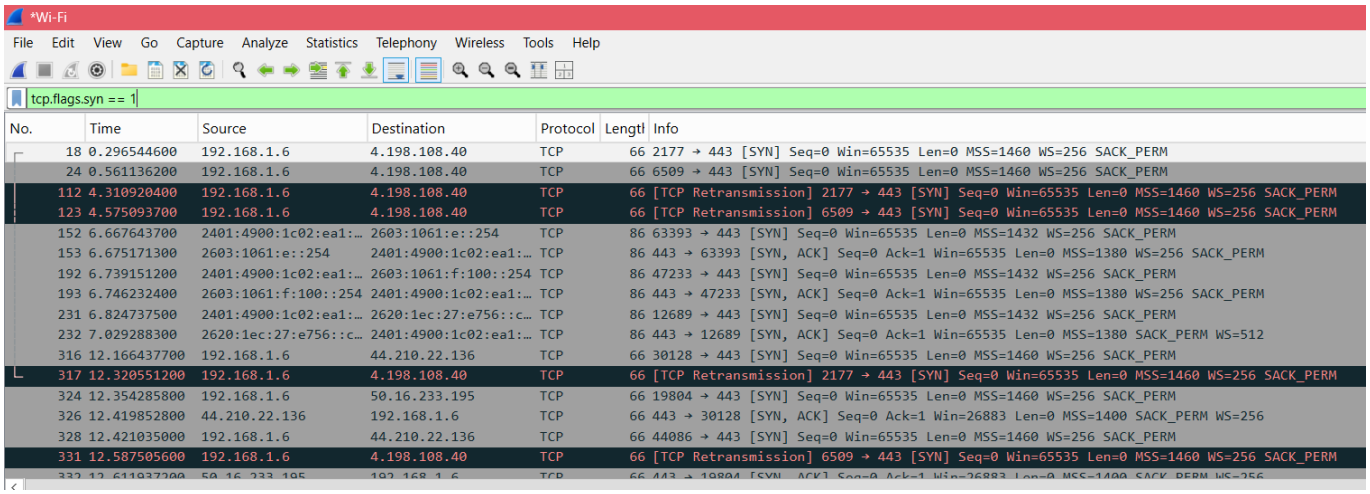
Step 4: Stop Capture and Apply Display Filter

1. Return to Wireshark.
2. Click the red square Stop icon to pause capture.
3. In the display filter bar, apply:

tcp.flags.syn == 1

4. Press Enter.

What this shows: All packets with the SYN flag set — these are the scan packets sent by Nmap.



Practical No: 03

Aim:

What is Port Scanning?

Port scanning is the process of probing a target host for open, closed, or filtered ports. It is used for:

- Network discovery
- Security auditing
- Vulnerability assessment
- Service identification

Port States

State	Meaning
Open	Service is listening and accepting connections
Closed	Port is reachable but no service is listening
Filtered	Firewall or filter is blocking probes; no response
Unfiltered	Port is accessible but state (open/closed) is unknown
Open|Filtered	Cannot determine if open or filtered

Setup

1. Open Zenmap (GUI for Nmap) or Command Prompt as Administrator.
2. Ensure you have permission to scan targets.
3. Use scanme.nmap.org (authorized) or your local network.

Scan 1: TCP ACK Scan

nmap -sA -T4 scanme.nmap.org

Explanation:

- ACK scan does NOT determine open/closed — it maps firewall rules.
- All 1000 ports are **unfiltered** (RST received) → no firewall blocking.
- **What it proves:** Target has no packet filtering on these ports.

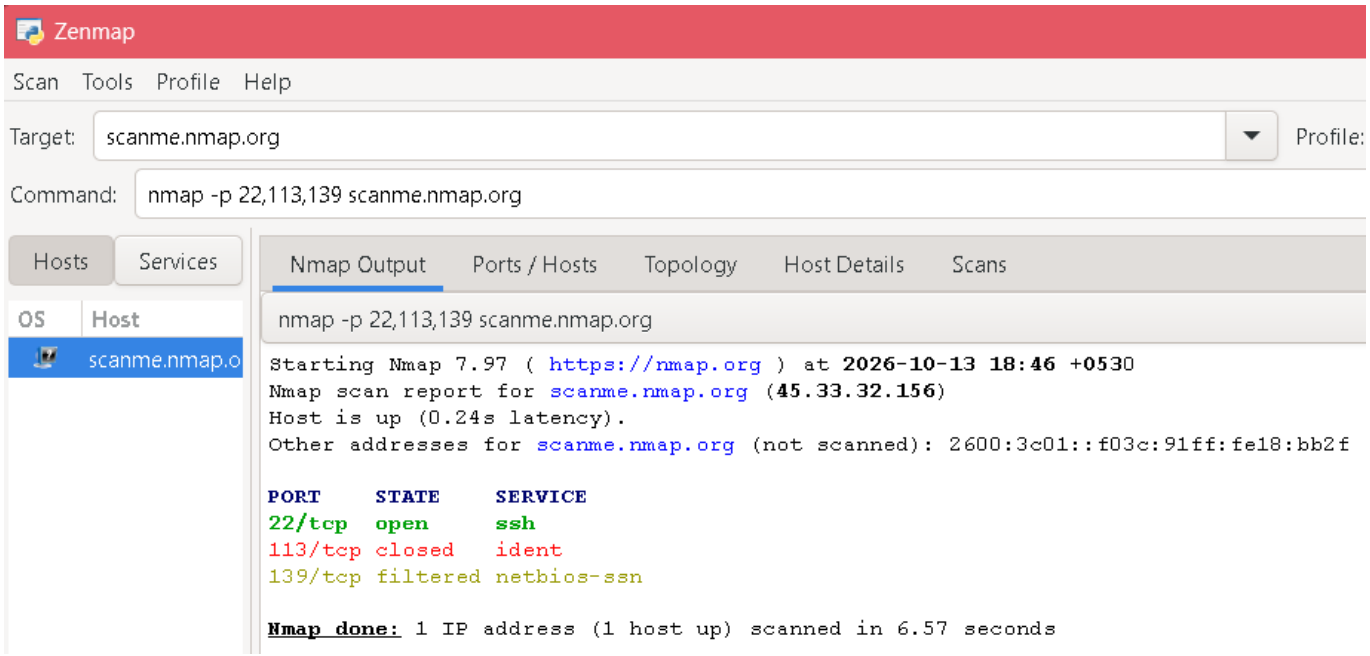


Scan 2: Specific Ports (SSH, Ident, NetBIOS)

nmap -p22,113,139 scanme.nmap.org

Explanation:

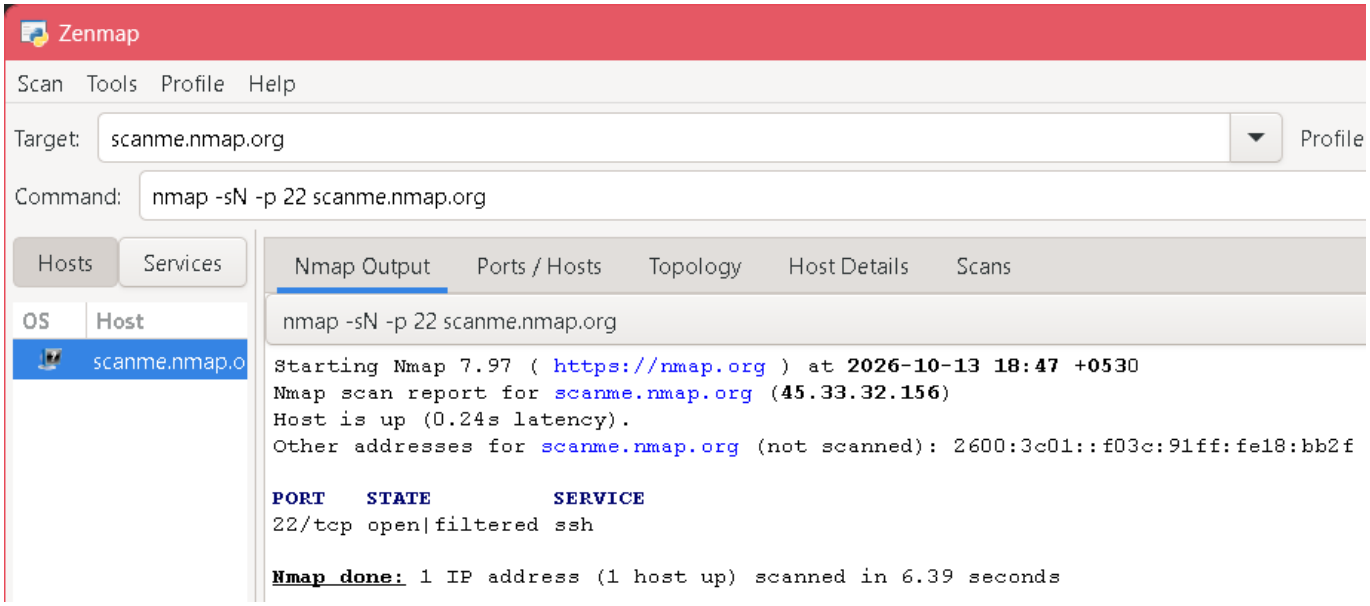
- **22/tcp open** → SSH service running.
- **113/tcp closed** → Port reachable, no service.
- **139/tcp filtered** → Firewall blocking.
- **What it proves:** Different port states can be identified.



Scan 3: TCP NULL Scan

nmap -sN -p 22 scanme.nmap.org

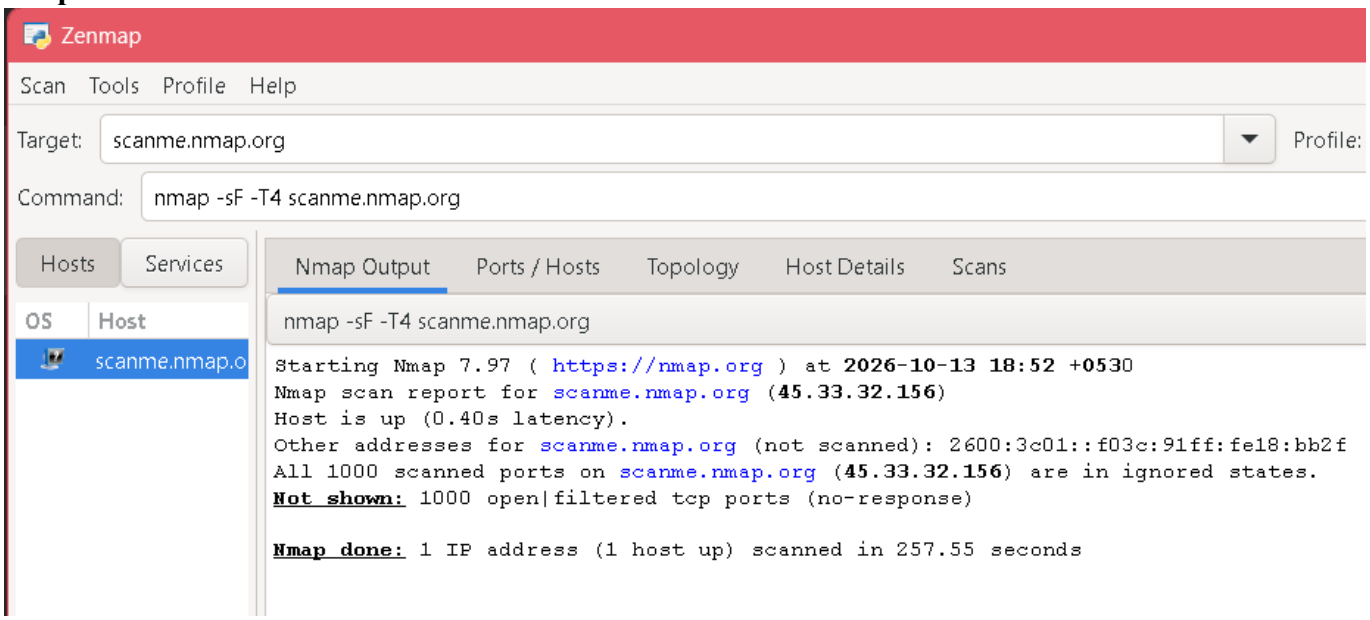
Output:



Scan 4: TCP FIN Scan

nmap -sF -T4 scanme.nmap.org

Output:



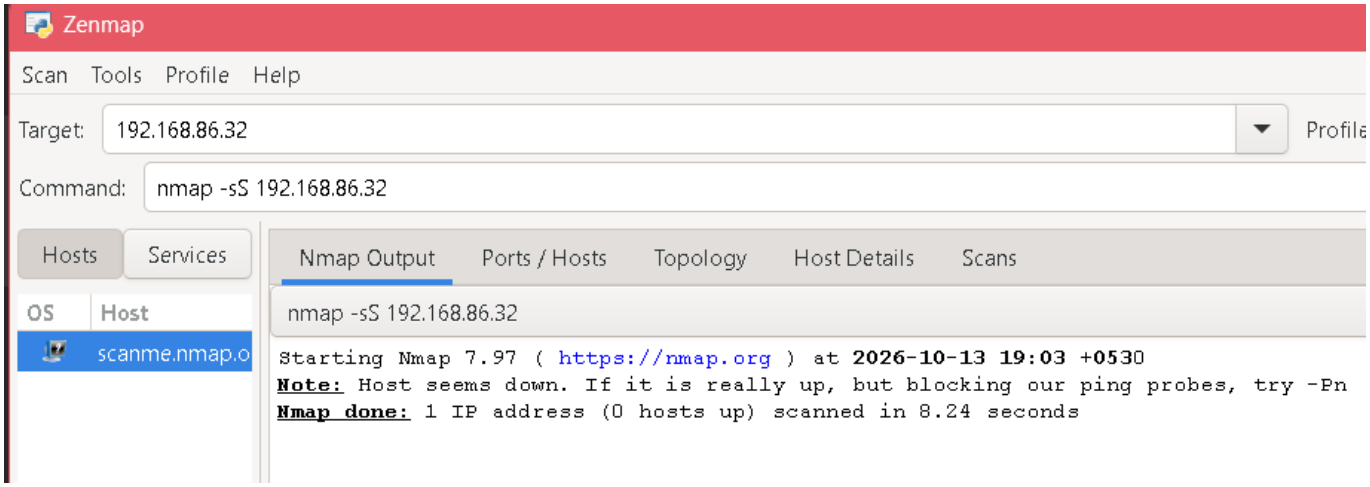
Explanation:

- FIN scan sends packet with **FIN flag only**.
- RST = closed. No response = open|filtered.
- All ports closed → all sent RST.
- **What it proves:** FIN scan can identify closed ports.

Scan 5: TCP SYN Scan on Local Host

nmap -sS 192.168.86.32

Output:



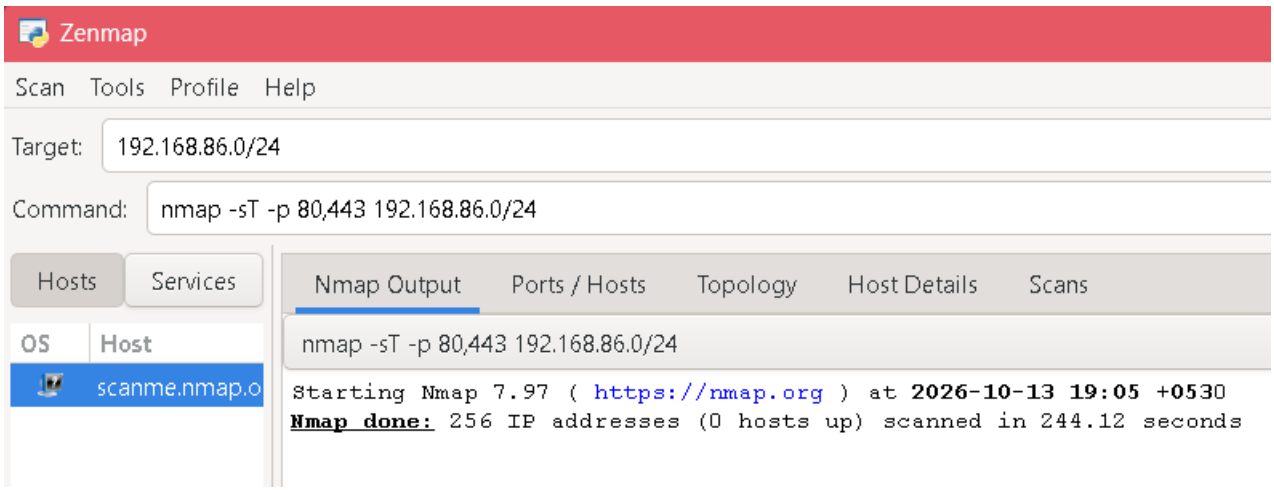
Explanation:

- Host did not respond to ping → Nmap assumes down.
- Use -Pn to skip ping and scan anyway.
- **What it proves:** Host discovery can fail if ICMP is blocked.

Scan 6: TCP Connect Scan on Subnet

nmap -sT -p 80,443 192.168.86.0/24

Output:



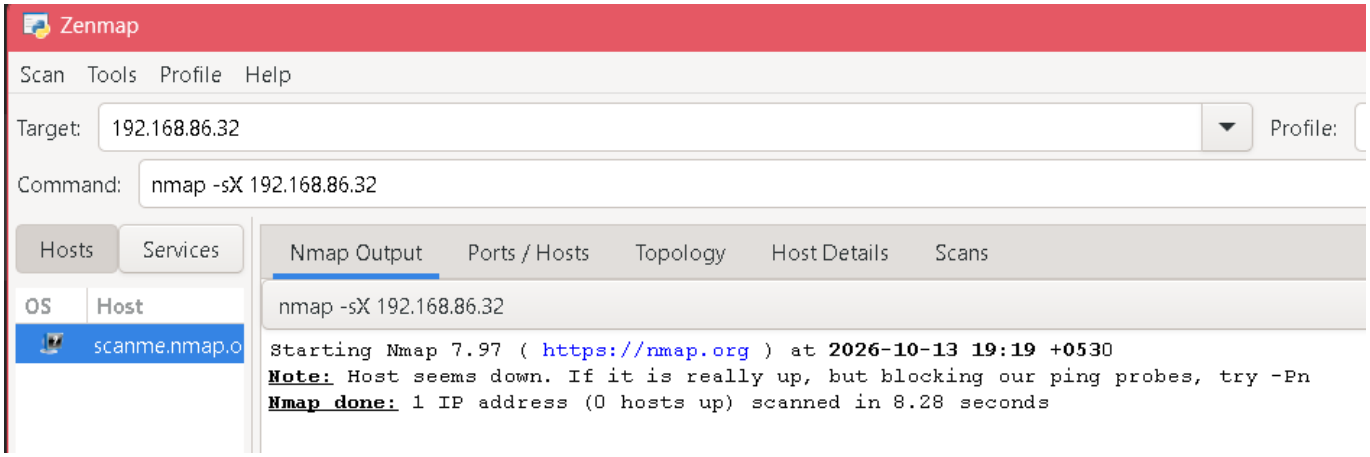
Explanation:

- Full TCP connect scan on 256 hosts.
- 0 hosts up → all hosts blocked ping or subnet is empty.
- **What it proves:** Subnet scanning can be slow and may miss hosts if ping is blocked.

Scan 7: TCP XMAS Scan

nmap -sX 192.168.86.32

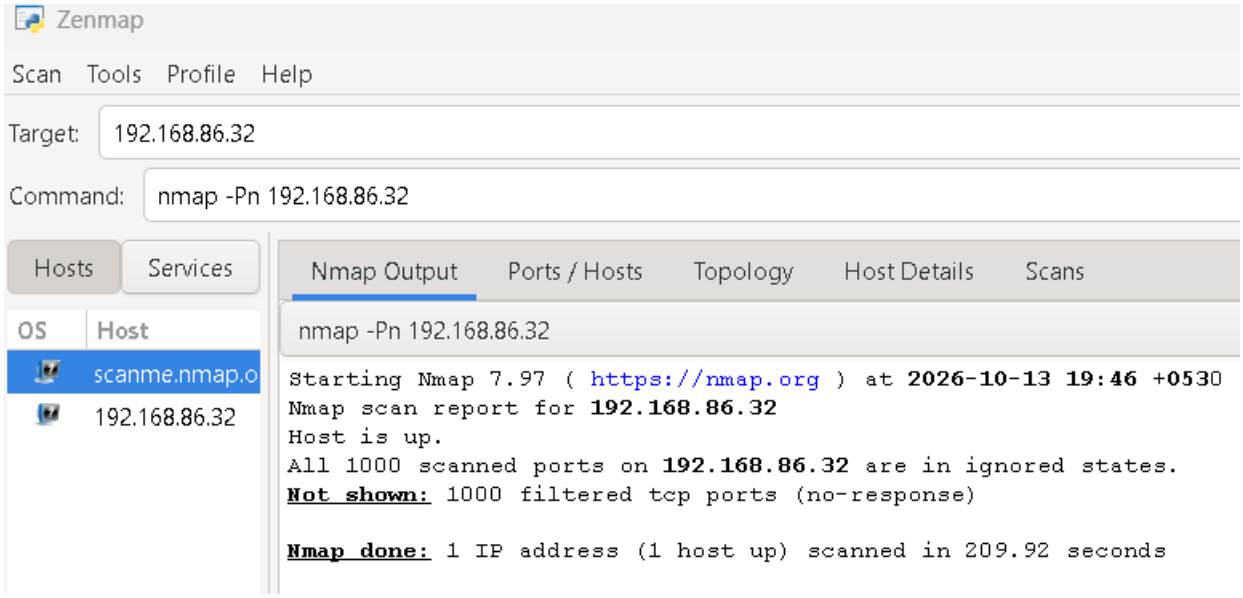
Output:



Followed by:

nmap -Pn 192.168.86.32

Output:



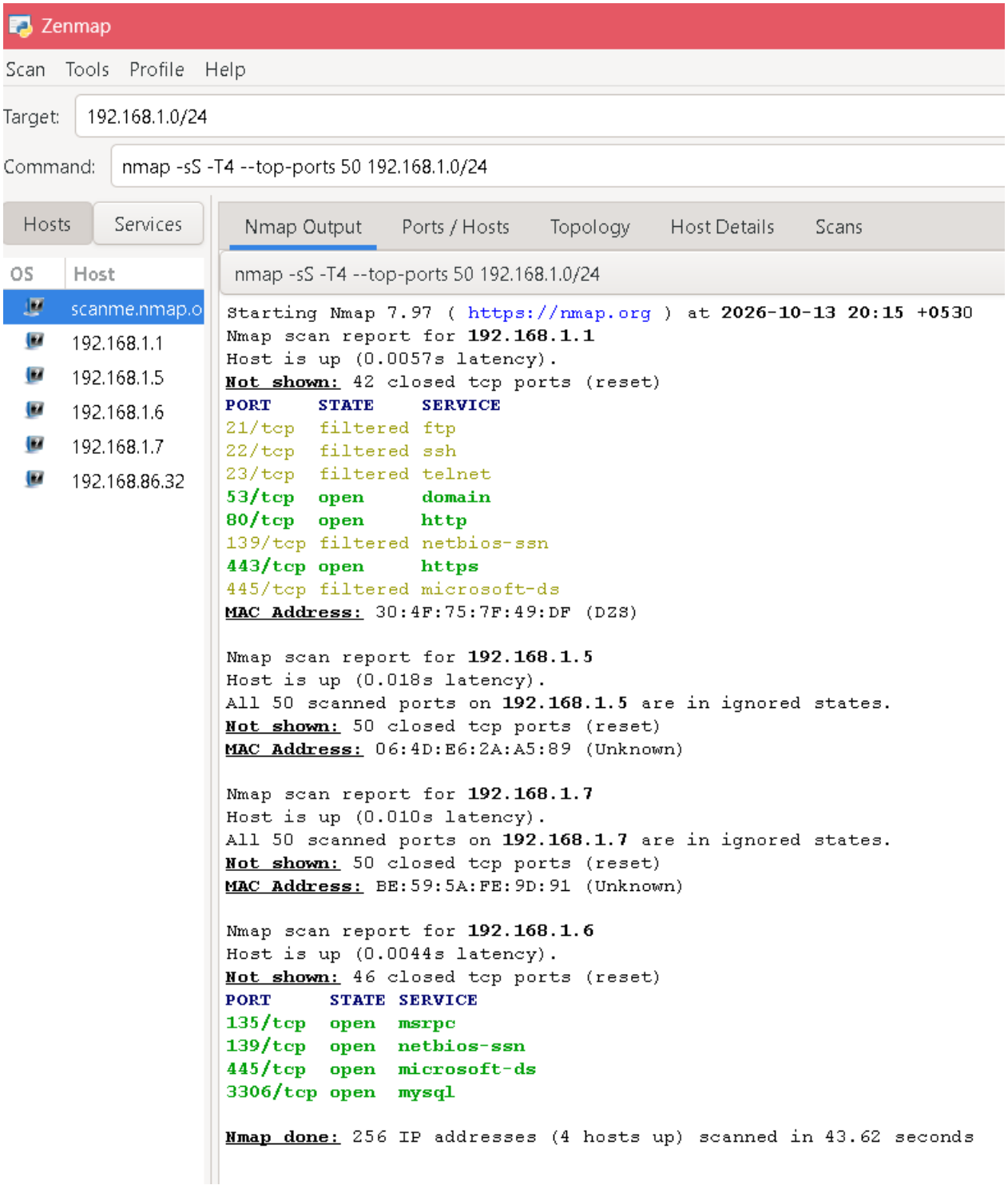
Explanation:

- XMAS scan sends FIN+PSH+URG.
- With -Pn, host is up but all ports filtered (no response).
- **What it proves:** XMAS scan can detect filtered ports.

Scan 8: Top 50 Ports on Local Network

nmap -sS -T4 --top-ports 50 192.168.1.0/24

Output:


Zenmap
Scan Tools Profile Help
Target: 192.168.1.0/24
Command: nmap -sS -T4 --top-ports 50 192.168.1.0/24
Hosts Services
OS Host
scanme.nmap.o
192.168.1.1
192.168.1.5
192.168.1.6
192.168.1.7
192.168.86.32
Nmap Output Ports / Hosts Topology Host Details Scans
nmap -sS -T4 --top-ports 50 192.168.1.0/24
Starting Nmap 7.97 (https://nmap.org) at 2026-10-13 20:15 +0530
Nmap scan report for 192.168.1.1
Host is up (0.0057s latency).
Not shown: 42 closed tcp ports (reset)
PORT STATE SERVICE
21/tcp filtered ftp
22/tcp filtered ssh
23/tcp filtered telnet
53/tcp open domain
80/tcp open http
139/tcp filtered netbios-ssn
443/tcp open https
445/tcp filtered microsoft-ds
MAC Address: 30:4F:75:7F:49:DF (DZS)
Nmap scan report for 192.168.1.5
Host is up (0.018s latency).
All 50 scanned ports on 192.168.1.5 are in ignored states.
Not shown: 50 closed tcp ports (reset)
MAC Address: 06:4D:E6:2A:A5:89 (Unknown)
Nmap scan report for 192.168.1.7
Host is up (0.010s latency).
All 50 scanned ports on 192.168.1.7 are in ignored states.
Not shown: 50 closed tcp ports (reset)
MAC Address: BE:59:5A:FE:9D:91 (Unknown)
Nmap scan report for 192.168.1.6
Host is up (0.0044s latency).
Not shown: 46 closed tcp ports (reset)
PORT STATE SERVICE
135/tcp open msrpc
139/tcp open netbios-ssn
445/tcp open microsoft-ds
3306/tcp open mysql
Nmap done: 256 IP addresses (4 hosts up) scanned in 43.62 seconds

Practical No: 04

Aim: **Perform Access Control List (Standard)**

What is an ACL?

An Access Control List (ACL) is a sequential list of permit/deny rules applied to a router interface to filter traffic. It controls whether packets are forwarded or blocked.

ACL Rules

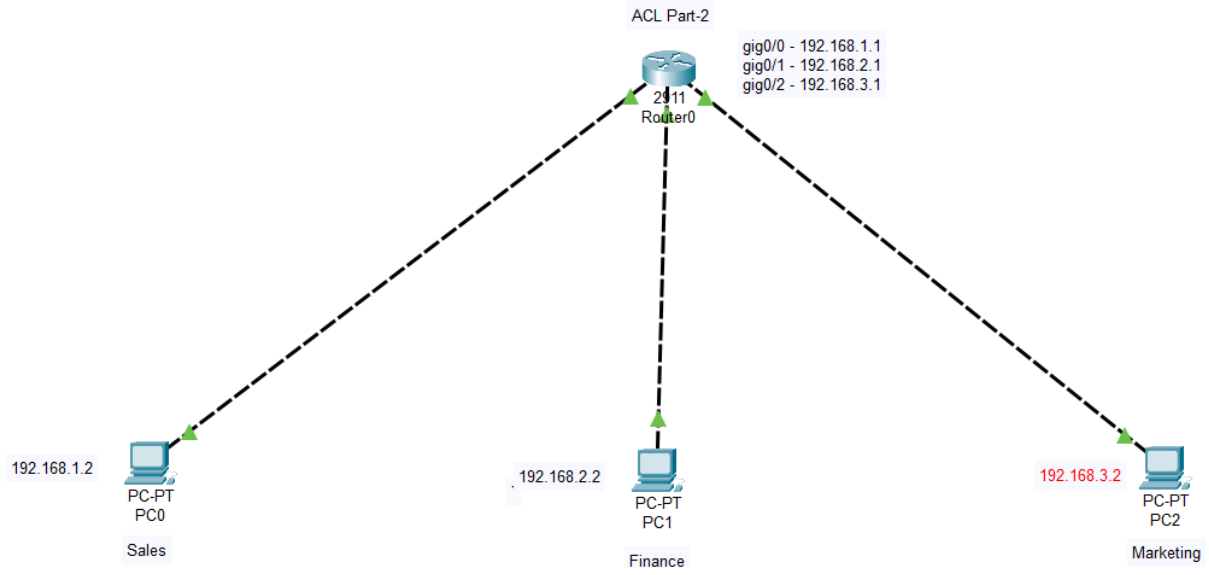
1. ACLs are processed **top-down**, first match wins.
2. There is an **implicit deny** at the end — if no rule matches, packet is dropped.
3. ACLs can be applied **in** (inbound) or **out** (outbound) on an interface.
4. Standard ACLs should be placed **close to the destination**.
5. Extended ACLs should be placed **close to the source**.

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Network
Router0	Gi0/0	192.168.1.1	255.255.255.0	Sales
Router0	Gi0/1	192.168.2.1	255.255.255.0	Finance
Router0	Gi0/2	192.168.3.1	255.255.255.0	Marketing
PC0	NIC	192.168.1.2	255.255.255.0	Sales
PC1	NIC	192.168.2.2	255.255.255.0	Finance
PC2	NIC	192.168.3.2	255.255.255.0	Marketing

Step 1: Build Topology

1. Place Router, PC0, PC1, PC2.
2. Connect all the devices
3. Configure IP addresses as per table above.



Step 2: Configure Router Interfaces

Router> enable

Router# configure terminal

Router(config)# interface GigabitEthernet0/0

Router(config-if)# ip address 192.168.1.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Router(config)# interface GigabitEthernet0/1

Router(config-if)# ip address 192.168.2.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Router(config)# interface GigabitEthernet0/2

Router(config-if)# ip address 192.168.3.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Step 3: Verify Connectivity (Before ACL)

From PC0:

PC> ping 192.168.2.2

PC> ping 192.168.3.2

Step 4: Configure Standard ACL

ACL (block only PC0):

Router(config)# access-list 1 deny host 192.168.1.2

Router(config)# access-list 1 permit any

Router(config)# interface GigabitEthernet0/0

Router(config-if)# ip access-group 1 in

Router(config-if)# exit

Step 5: Verify ACL (After ACL)

From PC0:

PC> ping 192.168.2.2

PC> ping 192.168.3.2

Expected: Both should fail (Request timed out).

From PC1:

PC> ping 192.168.3.2

Expected: Should succeed (PC1 is not blocked).

Step 6: Verify ACL on Router

Router# show access-lists

Router# show ip interface GigabitEthernet0/0

Router# show running-config

Expected Output:

Standard IP access list 1

10 deny host 192.168.1.2

20 permit any

Practical No: 05

Aim: **Configuring Local and Remote Access to Networking Devices**

What is Local Access?

Local access means connecting to a device **physically** via a console cable. It does not require network connectivity. It is used for initial configuration and disaster recovery.

What is Remote Access?

Remote access means connecting to a device **over the network** using protocols like Telnet or SSH.

Key Concepts

- **VTY (Virtual Teletype):** Virtual lines for remote access.
- **RSA Keys:** Encryption keys used by SSH.
- **Domain Name:** Required for SSH key generation (e.g., mac.com).
- **Enable Secret:** Encrypted password for privileged mode.
- **Login Local:** Uses local username/password database.

PART A: Configuration 1 — SSH on Router

Topology



Step 1: Configure Router Hostname and Domain

Router> enable

Router# configure terminal

Router(config)# hostname r1

r1(config)# ip domain-name mac.com

Step 2: Generate RSA Keys

r1(config)# crypto key generate rsa

The name for the keys will be: r1.mac.com

Choose the size of the key modulus: [512]: 2048

Step 3: Create Username and Password

r1(config)# username admin password pass

r1(config)# enable secret pass

Step 4: Configure SSH (VTY Lines)

r1(config)# line vty 0 1

r1(config-line)# login local

r1(config-line)# transport input ssh

r1(config-line)# ip ssh version 2

r1(config-line)# exit

r1(config)# exit

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname r1
r1(config)#ip domain-name mac.com
r1(config)#crypto key generate rsa
The name for the keys will be: r1.mac.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

r1(config)#username admin password passwd
*Mar 1 0:1:49.290: %SSH-5-ENABLED: SSH 1.99 has been enabled
r1(config)#enable secret pass
r1(config)#line vty 0 1
r1(config-line)#login local
r1(config-line)#transport input ssh
r1(config-line)#ip ssh version 2
r1(config)#exit
r1#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 5: Test SSH from PC

From PC0 Command Prompt:

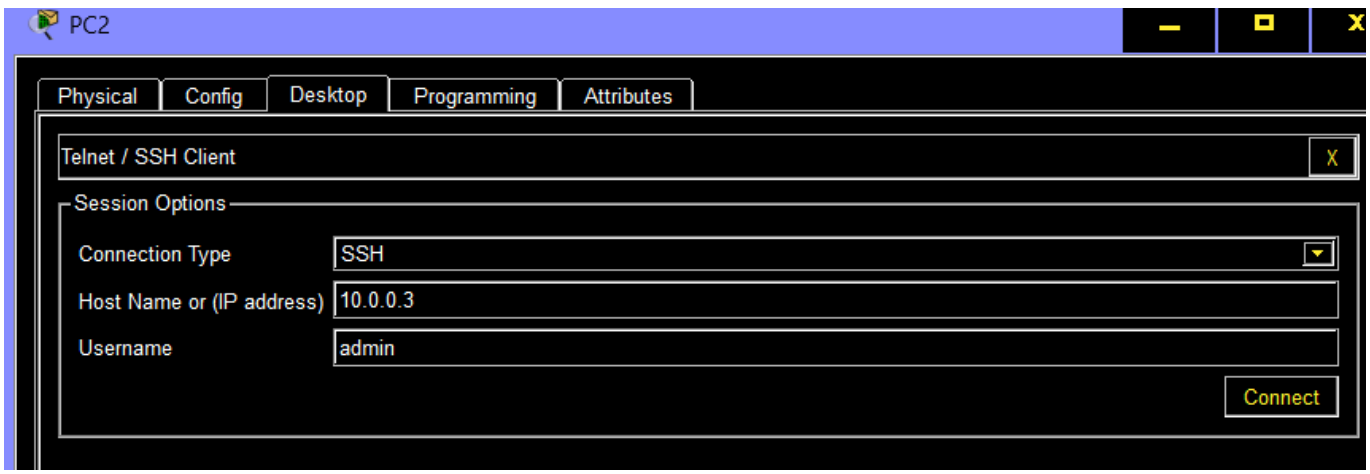
PC> ssh -l admin 10.0.0.3

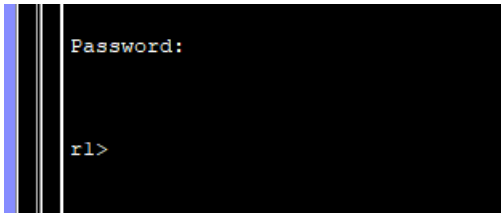
Password: pass

r1> enable

Password: pass

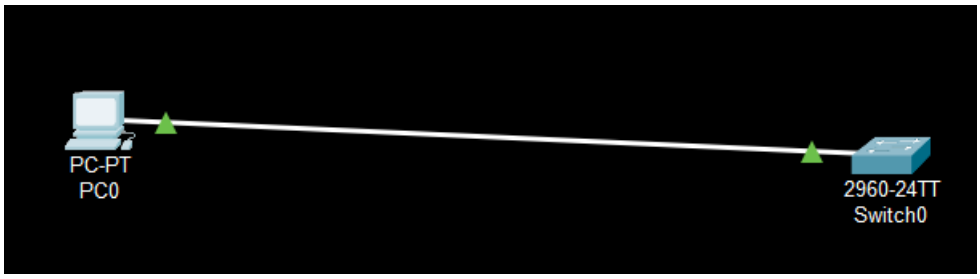
r1#





PART C: Configuration 2 — Telnet on Switch

Topology



Step 1: Configure PC

- IP: 10.0.0.1
- Subnet Mask: 255.0.0.0

Step 2: Configure Switch Basic Settings

Switch> enable

Switch# configure terminal

Switch(config)# enable password password

Step 3: Configure VLAN 1 (Management IP)

text

Switch(config)# interface vlan 1

Switch(config-if)# ip address 10.0.0.5 255.0.0.0

Switch(config-if)# no shutdown

Switch(config-if)# exit

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#enable password passwd
Switch(config)#interface vlan 1
Switch(config-if)#ip address 10.0.0.5 255.0.0.0
Switch(config-if)#no shutdown
```

Step 4: Configure Telnet (VTY Lines)

Switch(config)# line vty 0 15

Switch(config-line)# password pass

Switch(config-line)# login

Switch(config-line)# exit

Switch(config)# exit

```
Switch(config-line)#password passwd
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 5: Test Telnet from PC

PC> telnet 10.0.0.5

Password: pass

Switch>

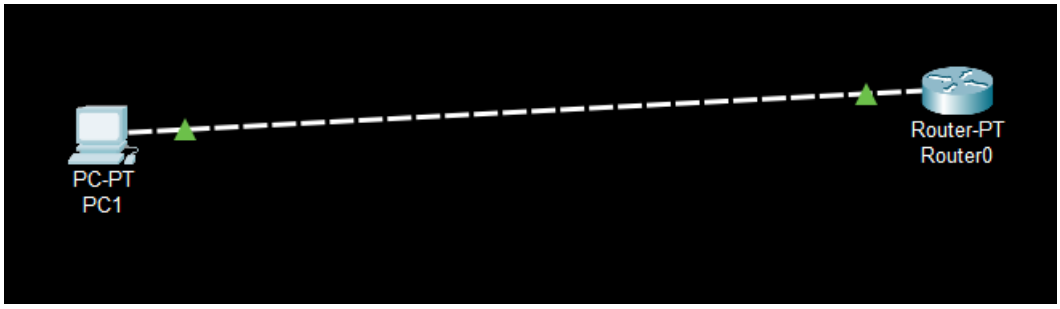
```
C:\>telnet 10.0.0.5
Trying 10.0.0.5 ...Open


User Access Verification

Password:
Switch>
```

PART D: Configuration 3 — Telnet on Router

Topology



Step 1: Configure PC

- IP: 10.0.0.2
- Subnet Mask: 255.0.0.0
- Gateway: 10.0.0.3

Step 2: Configure Router Basic Settings

Router> enable

Router# configure terminal

Router(config)# username admin password pass

Router(config)# enable secret pass

Step 3: Configure Telnet (VTY Lines)

Router(config)# line vty 0 1

Router(config-line)# login local

Router(config-line)# transport input telnet

Router(config-line)# exit

Router(config)# exit

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#username admin password passwd
Router(config)#enable secret passwd
Router(config)#line vty 0 1
Router(config-line)#login local
Router(config-line)#transport input telnet
Router(config-line)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4: Test Telnet from PC

PC> telnet 10.0.0.3

Username: admin

Password: pass

```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 10.0.0.3
Trying 10.0.0.3 ...Open


User Access Verification

Username: passwd
Password:
% Login invalid

Username: admin
Password:
Router>
```

Practical No: 06

Aim: **Configuring DHCP**

What is DHCP?

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and other network configuration (subnet mask, gateway, DNS) to devices on a network. It uses **UDP ports 67 (server) and 68 (client)**.

Why DHCP?

- Manual IP configuration is error-prone and slow.
- DHCP eliminates IP conflicts.
- Centralized management of IP addresses.
- Essential for large networks, Wi-Fi, and guest networks.

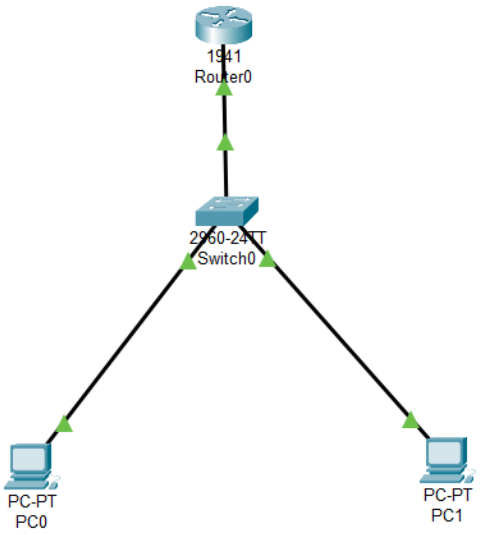
Topology

IP Configuration Table

Device	Interface	IP Address	Role
Router0	Gi0/0	10.0.0.3	Gateway + DHCP Server
PC0	NIC	DHCP (10.0.0.1)	Client
PC1	NIC	DHCP (10.0.0.2)	Client

Step 1: Build Topology

1. Add Router0 (1941), Switch0 (2960), PC0, PC1.
2. Connect Router0 Gi0/0 to Switch0 Fa0/1.
3. Connect PC0 to Switch0 Fa0/2, PC1 to Switch0 Fa0/3.



Step 2: Configure Router Gateway Interface

Router> enable

Router# configure terminal

Router(config)# interface GigabitEthernet0/0

Router(config-if)# ip address 10.0.0.3 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit

Step 3: Configure DHCP Pool and Exclusions

Router(config)# ip dhcp excluded-address 10.0.0.3

Router(config)# ip dhcp pool msc

Router(dhcp-config)# network 10.0.0.0 255.255.255.0

Router(dhcp-config)# default-router 10.0.0.3

Router(dhcp-config)# dns-server 10.0.0.3

Router(dhcp-config)# exit

Router(config)# exit

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 10.0.0.3 255.0.0.0
Router(config-if)#no shutdown

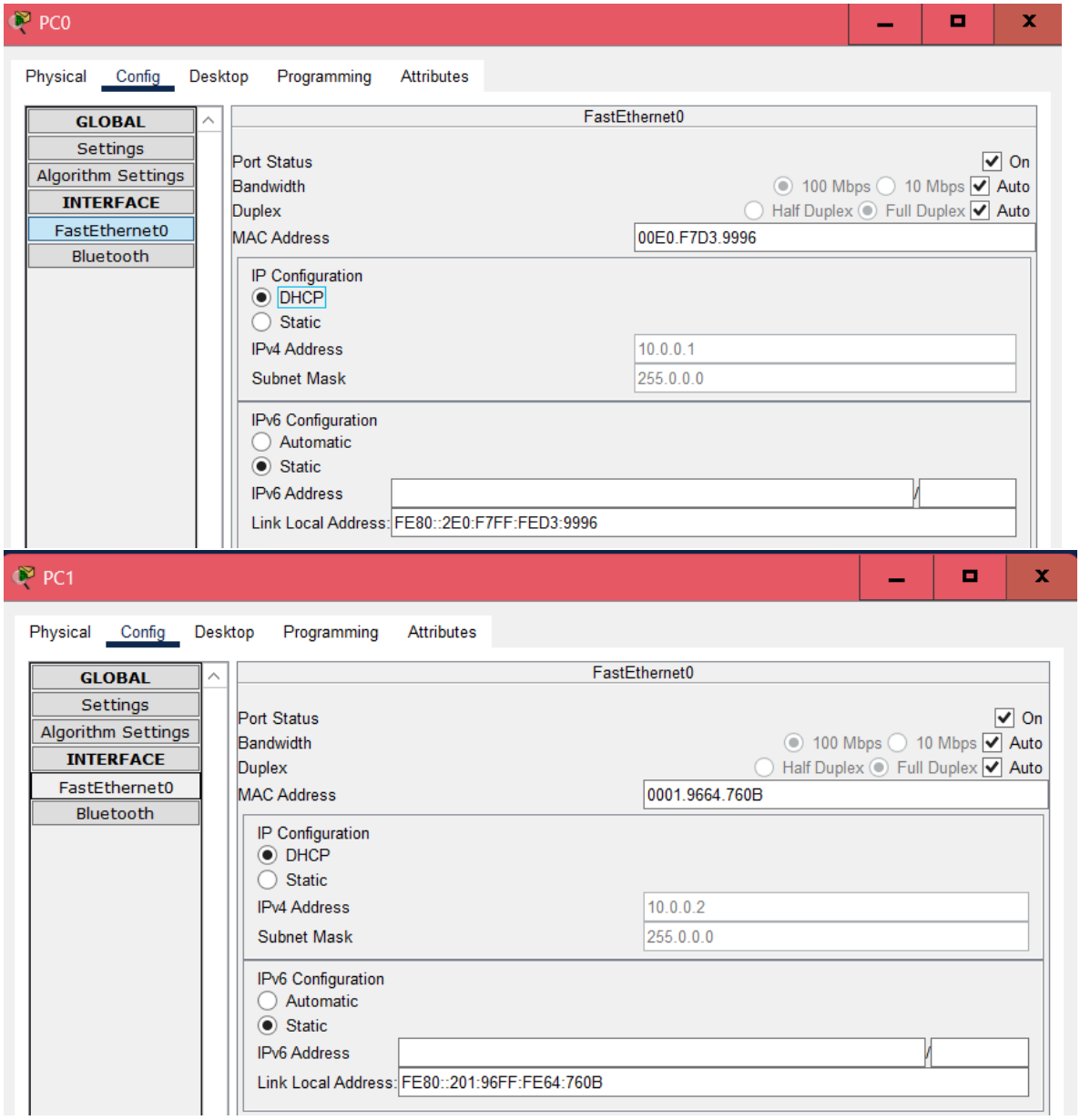
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 10.0.0.3
Router(config)#ip dhcp pool msc
Router(dhcp-config)#network 10.0.0.0 255.0.0.0
Router(dhcp-config)#default-router 10.0.0.3
Router(dhcp-config)#dns-server 10.0.0.3
Router(dhcp-config)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 4: Configure PCs to Use DHCP

1. Click PC0 → Desktop → IP Configuration.
2. Select DHCP radio button.
3. Wait for "DHCP request successful."
4. Repeat for PC1.



Step 5: Verify on Router

Router# show ip dhcp binding

Step 6: Verify on PC

PC> ipconfig /all

Practical No: 07

Aim: **Performing Backup and Restore of Running Configuration on a Cisco Router**

What is Backup and Restore?

- **Backup:** Copying the router's configuration to an external server (TFTP) for safekeeping.
- **Restore:** Copying the saved configuration back to the router from the TFTP server.

Why Backup?

- Router failure or factory reset
- Accidental configuration changes
- Hardware replacement
- Disaster recovery
- Compliance and auditing

What is TFTP?

TFTP (Trivial File Transfer Protocol) is a simple, connectionless file transfer protocol. It uses **UDP port 69**. It is lightweight and commonly used for:

- Router/switch config backup
- IOS image transfer
- Firmware updates

Topology

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Role
Router0	Fe0/0	10.0.0.1	255.0.0.0	Router
Server0	NIC	10.0.0.2	255.0.0.0	TFTP Server

Step 1: Build Topology

1. Add Router0 (2811) and Server0.
2. Connect Router0 FastEthernet0/0 to Server0
3. Configure IP addresses as per table above.

Step 2: Configure Router

Router> enable

Router# configure terminal

Router(config)# interface FastEthernet0/0

Router(config-if)# ip address 10.0.0.1 255.0.0.0

Router(config-if)# no shutdown

Router(config-if)# exit

Router(config)# hostname R1

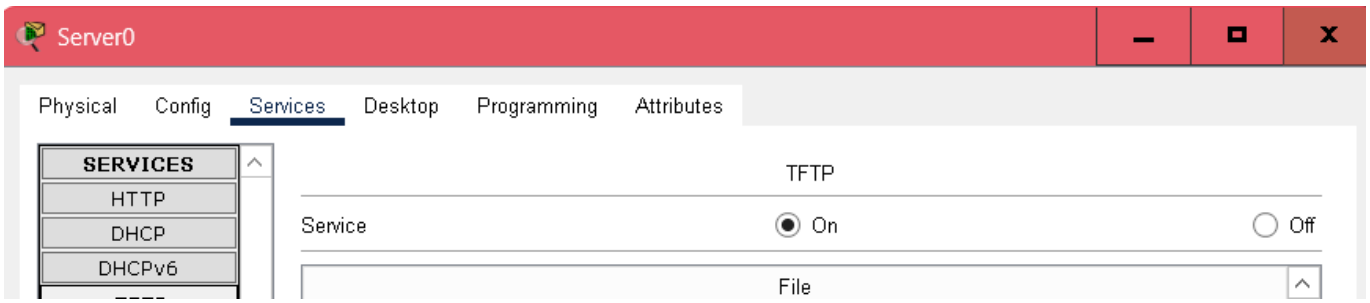
R1(config)# exit

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0
                                ^
% Invalid input detected at '^' marker.

Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#hostname R1
R1(config)#exit
R1#
```

Step 3: Configure TFTP Server

1. Click Server0 → Desktop → IP Configuration.
2. Set IP: 10.0.0.2, Mask: 255.0.0.0.
3. Go to Services → TFTP → Ensure TFTP is **On**.



[Insert server config screenshot]

Step 4: Test Connectivity

From Router:

R1# ping 10.0.0.2

Step 5: Backup Running Config to TFTP

R1# copy running-config tftp

Address or name of remote host []? 10.0.0.2

Destination filename [R1-confg]? Backup2026

Writing running-config...!!

[OK - 608 bytes]

608 bytes copied in 0 secs

R1#

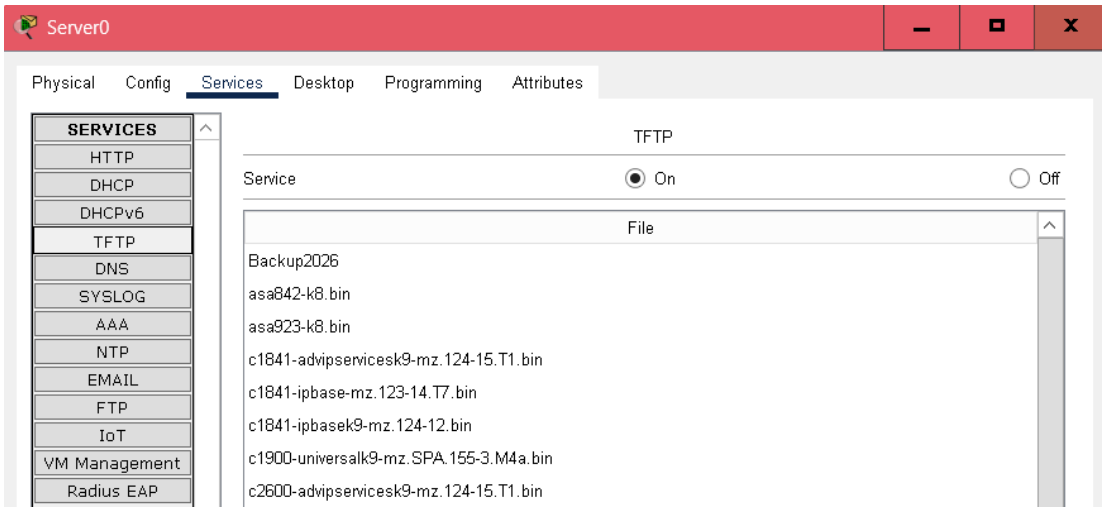
```
R1#copy running-config tftp
Address or name of remote host []? 10.0.0.2
Destination filename [R1-confg]? Backup2026

Writing running-config...!!
[OK - 688 bytes]

688 bytes copied in 0 secs
R1#
```

Step 6: Verify Backup on Server

1. Click Server0 → Services → TFTP.
2. You should see Backup2026 in the file list.



Step 7: Simulate Failure (Erase Config)

R1# erase startup-config

R1# reload

Step 8: Restore Config from TFTP

Router# copy tftp running-config

Address or name of remote host []? 10.0.0.2

Source filename []? Backup2026

Destination filename [running-config]?

Accessing tftp://10.0.0.2/Backup_2026...

Loading Backup_2026 from 10.0.0.2: !

[OK - 608 bytes]

608 bytes copied in 0 secs

```
R1>enable
R1#copy tftp running-config
Address or name of remote host []? 10.0.0.2
Source filename []? Backup2026
Destination filename [running-config]?

Accessing tftp://10.0.0.2/Backup2026....
Loading Backup2026 from 10.0.0.2: !
[OK - 688 bytes]

688 bytes copied in 3.014 secs (228 bytes/sec)
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 9: Verify Restore

Router# show running-config

R1# show ip interface brief

```
R1#show running-config
Building configuration...

Current configuration : 728 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!

R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.0.1        YES manual up                    up
FastEthernet1/0        unassigned      YES NVRAM  administratively down down
Serial2/0              unassigned      YES NVRAM  administratively down down
Serial3/0              unassigned      YES NVRAM  administratively down down
FastEthernet4/0        unassigned      YES NVRAM  administratively down down
FastEthernet5/0        unassigned      YES NVRAM  administratively down down
R1#
```